NAPG: Neighborhood-Assisted Multiprototype Group Model for Cross-Domain Semantic Segmentation of Remote Sensing Images

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***Abstract*— Unsupervised domain adaptation (UDA) is crucial for conciseness and readability (RS-SS), particularly when data distributions differ between source and target domains. Existing prototype-based UDA methods struggle with complex land cover class distributions and spatial information capture. To address these limitations, the neighborhood-assisted prototype group (NAPG) model is proposed. This model enhances cross-domain adaptability and spatial context richness by dynamically determining the number of prototype features and integrating neighborhood similarity gradients. Specifically, NAPG employs the cross-domain representation of multiprototype group (CDR-MPG) module to generate multiprototype group (MPG), capturing land cover complexity more effectively. Additionally, the gradient neighborhood consistency estimation (GNCE) module improves spatial representation by reducing intraclass variance and alleviating feature inconsistency. Experiments demonstrate that the proposed NAPG model outperforms the state-of-the-art UDA methods across multiple datasets, achieving a mean intersection over union (mIoU) improvement of 3%. The source code is publicly available at https://github.com/Fanrongbo/NAPG-UDA-RS-SS**

***Index Terms*— Gradient neighborhood consistency estimation (GNCE), multiprototype group (MPG), remote sensing image semantic segmentation, unsupervised domain adaptation (UDA).**

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I. Introduction

CONCISENESS and readability (RS-SS) is a crucial Earth observation technique that involves analyzing remote sensing data to assign precise labels to each pixel [1]. Recent advances in deep learning for RS-SS, coupled with the growing availability of public datasets [2], [3], [4], have greatly improved RS-SS performance. However, different RS-SS datasets exhibit varying spatial resolution, spectral radiance, and noise level, which can lead to covariate shifts [5] that adversely impact model performance. Specifically, models trained in a supervised manner on labeled datasets (referred to as the source domain) often struggle to generalize to other unlabeled datasets (referred to as the target domain). This limitation necessitates a costly manual labeling process for new data. This process is not only resource-intensive but also prevents the reuse of the existing labeled datasets and hampers the rapid deployment of models for urgent Earth monitoring tasks.

Unsupervised domain adaptation (UDA) is a crucial technique for enhancing model performance on unlabeled target domain data. It achieves this by learning the mapping relationship or feature representation between the source and target domains, thereby enabling models to transfer knowledge from labeled source domain data to improve performance on the target domain without requiring target domain labels. UDA methods can be broadly categorized into two types: marginal distribution alignment and conditional distribution alignment [6]. Marginal distribution alignment reduces overall distributional differences between domains. Adversarial learning is the most prominent UDA method. These algorithms use generative adversarial training to produce common features across domains using original images [1], [2], [7], [8], [9], [10], predicted probabilities [11], [12], predicted entropy values [13], [14], [15], or prior class histograms [16], [17]. In contrast, conditional distribution alignment focuses on reducing discrepancies in class-specific distributions. A core element is the development of reliable, high-confidence pseudo-label selection methods [8]. An effective UDA strategy often combines marginal and conditional distribution alignment [1]. However, these methods face significant challenges. On the one hand, marginal distribution alignment lacks label constraints, which may lead to the neglect of small or complex land cover classes. On the other hand, traditional conditional distribution

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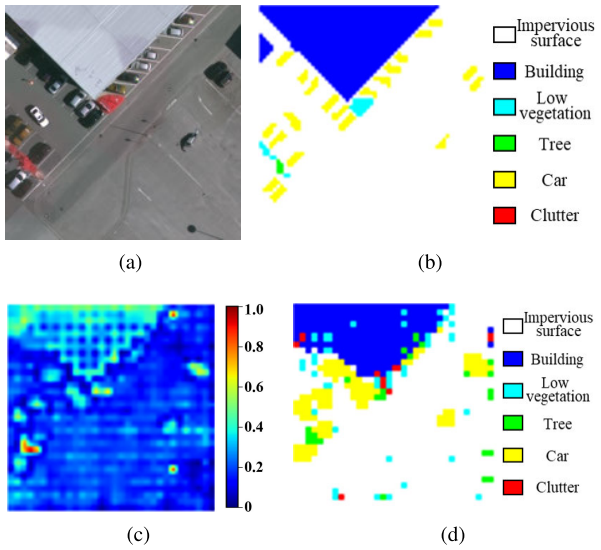


Fig. 1. Distribution of grid-like artifacts and misclassified points in prototype similarity prediction. $\mathcal{I}^{\mathcal{T}}$ is target-domain image. $\mathcal{Y}^{\mathcal{T}}$ is target–domain label. $\hat{y}^{\mathcal{T}}_{\max}$ is the maximum prediction probability of target domain sample. $O^{\mathcal{T}}$ is the prediction result of target-domain sample. (a) $\mathcal{I}^{\mathcal{T}}$. (b) $\mathcal{Y}^{\mathcal{T}}$. (c) $\hat{y}^{\mathcal{T}}_{\max}$. (d) $\hat{y}^{\mathcal{T}}$.

alignment involves pixel-level feature alignment, resulting in high computational costs.

The prototype feature is a compact and robust representation with low computational complexity [18]. These features are generated by clustering distinct classes within the embedding space and enabling pixelwise classification based on the similarity between features and their corresponding prototypes. In the context of UDA, prototype features help establish a stable shared feature space between the source and target domains. By reducing the distribution differences between target-domain features and prototype features [1], [11], [18], [19], [20], [21], domain alignment is facilitated, and constraints are imposed on the cross-class deep feature distributions. However, prototype-based UDA models for RS-SS still face some challenges.

1) A single prototype is often insufficient to capture the high intraclass variability of certain land cover objects in remote sensing images. Conversely, employing multiple prototypes for classes with low feature dispersion can introduce redundancy and lead to interclass feature mismatches.
2) Lack of spatial information in prototype features: due to the lack of spatial information in prototype features, it is challenging to effectively capture the spatial context of neighboring regions, which leads to errors in pixel-level predictions based on similarity measurements. For example, the model may misclassify discrete noise points in neighboring regions and even produce a grid-like artifacts in the prediction results [22], as shown in Fig. 1. When relying solely on similarity measurements, although the classification results of pixels are influenced by surrounding pixels, this influence is not fully utilized in the absence of spatial information.

To address the above challenges, a neighborhood-assisted prototype group (NAPG) model for cross-domain RS-SS is proposed. The NAPG model is specifically designed for RS-SS UDA tasks that involve significant distributional differences between land cover classes. Unlike the existing prototype-based UDA approaches, the NAPG model adaptively determines the number of prototype features based on the distributional dispersion of different land cover classes, forming a multiprototype group (MPG). By incorporating auxiliary spatial information through neighborhood consistency estimation (GNCE), it mitigates prototype matching errors caused by the lack of spatial information. The main contributions of this study are as follows.

1) Based on the feature distribution characteristics of different land cover categories in the source and target domains, the optimal number of prototype features for each category is determined to form an MPG. The proposed MPG-based cross-domain representation of multi-prototype group (CDR-MPG) module enables efficient and accurate representation of feature distributions, thereby enhancing the performance of UDA for RS-SS.
2) By embedding the GNCE module, the spatial topological information in the sparse MPG is compensated. This improves the spatial smoothness and consistency of MPG similarity prediction.
3) The CDR-MPG module and GNCE module are integrated into the widely used RS-SS model, combined with UDA optimization strategies designed for RS-SS.

This article is organized as follows. Section II reviews the related work. Section III provides a detailed description of the proposed UDA module and UDA optimization strategies. Section IV introduces the experimental setup, performance analysis of the MPG, GNCE performance analysis, comparison with current state-of-the-art algorithms, and visualization results. Section V summarizes this article. Section VI is the appendix, proving the advantages of the prototype similarity cross-entropy optimization function and expanding the performance comparison experiments.

II. Related Work

A. UDA Based on Semantic Segmentation

UDA is a technique that transfers a model from a source domain to a target domain with different data distributions but consistent underlying classes. By minimizing the distribution gap, UDA improves model performance on unlabeled target-domain data. UDA methods focus on aligning conditional and marginal distributions and can be categorized into feature-level, raw image-level, and decision-level alignment.

Feature-level alignment is a fundamental method in UDA, encompassing domain feature discrepancy alignment and domain adversarial training. Early UDA approaches typically aligned features by quantifying domain feature discrepancies across various feature levels. Common discrepancy metrics include Wasserstein distance [23], CORAL [24], JMMD [25], covariance regularization [26], and optimal transport-based alignment [11], [27]. These methods can achieve conditional and marginal distribution alignment, depending on whether features are aligned per class. While these methods work well

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for fine-grained feature alignment, they tend to be computationally expensive, prone to perturbations, and fail to fully account for spatial topological information when dealing with pixel-level features. In domain adversarial training strategies, Vu et al. [14] and Saito et al. [28] were the pioneers in incorporating adversarial training into UDA, using it to progressively learn the shared features between the source and target domains. Luo et al. [29] introduced adaptive adversarial loss weighting to enhance conditional distribution alignment. Xu et al. [30] proposed a class-aware generative adversarial network to improve inter- and intraclass consistency through adaptive class selection. Zhang et al. [31] employed curriculum learning to align local and global spatial features. Combined strategies [32], [33] integrate domain adversarial and discrepancy-based alignment for better performance on fine-grained tasks.

Raw image level focuses on marginal distribution alignment, reducing visual discrepancies through data augmentation and style transfer. For data augmentation, Shen et al. [9] proposed cross-domain mixed augmentation, mixing source samples with target style versions to improve feature learning. Ran et al. [34] used domain fusion images to enhance self-training. In style transfer, Cai et al. [10] applied GANs for image translation, and Liang et al. [1] used CycleGAN for heterogeneous data transformation, ensuring instance-level alignment through cycle consistency. Chen et al. [8] combined data augmentation and style transfer using a dual-path CycleGAN, performing class-level stitching to address domain shifts. Other studies [2], [13] incorporated data augmentation to improve generalization.

Decision-level alignment relies on pseudo-labeled target domain samples for conditional distribution alignment. High-confidence pseudo-label selection strategy [35] is key, with strategies based on probability thresholds [8], entropy thresholds [5], [36], [37], entropy ranking [10], [38], and prototype feature distance thresholds [9], [18], [39], [40], [41]. Iqbal and Ali [16] improved pseudo-label accuracy by integrating prior information. Xie et al. [36] proposed a region selection strategy that combines region impurity and prediction uncertainty. He et al. [42] modeled noise in pseudo-labels using class-related noise matrices, while Liu et al. [43] introduced multilevel pseudo-label correction at class, pixel, and superpixel levels. Combined methods [1], [2], [44] improve adaptation and generalization by addressing multiple levels of domain alignment.

B. UDA Based on Prototype Features

Prototype features describe the central tendency of feature distributions for a given object class. They are computed by aggregating features from the samples of the same class, providing a summary of the overall class characteristics. In prototype classification, the distance between the query feature (deep features of target domain samples) and the prototypes of each class is measured. In UDA, prototype-based methods are effective. They allow pseudo-labeling of target-domain samples by comparing their distance to class prototypes and are robust to discrete feature distributions. This approach reduces the distribution gap between target- and source-domain features, facilitating UDA.

Deng et al. [45] proposed a method that combines clustering learning with discriminative feature alignment, systematically demonstrating how UDA can be achieved using prototype features. Xu et al. [11] addressed the problem of traditional optimal transport methods, which perform coarse-grained matching while ignoring domain-specific structures, by using single prototype information and intradomain structures. Chen et al. [40] used class prototypes to classify features between the source and target domains and guide domain alignment for specific classes, while also improving prototype representation through increased intraclass consistency and interclass separability.

To dynamically update prototype features during model optimization and improve their association with deep features, some methods adjust prototypes using each batch of training data, gradually adapting to the target-domain distribution [1], [20], [39]. Zhu et al. [18] proposed a domain-level prototype memory module to store and update pseudo-invariant features of each class in both the source and target domains.

To overcome the limitations of single prototype representation and enhance its feature representation capability, the existing methods have proposed various multiprototype feature generation and optimization strategies for different task scenarios. Ning et al. [19] used K-means clustering to generate multiprototype features, determined the prototype number to be ten through hyperparameters, and optimized it with soft alignment loss; Gao et al. [44], Zhou et al. [46], and Gao et al. [47] all adopted online clustering to generate multiprototype features by combining random initialization with the Sinkhorn–Knopp algorithm and progressively improved model performance. Li et al. [48] generated multiprototype features using manually designed text prompts for semi-supervised domain adaptation tasks; Liao et al. [49] enhanced multiprototype features to generate multiprototype features and optimized them with uncertainty-weighted loss. However, most of the above methods rely on hyperparameters to determine the number of prototypes, which makes their adaptability to different tasks and datasets relatively weak.

To address the lack of spatial context in prototype-based methods, researchers have explored incorporating spatial information to improve model performance. Shen et al. [50] and Zhang and Sato [51] used superpixel clustering, but these methods are computationally expensive and neglect spatial topological information. Dong et al. [52] improved boundary recognition through online clustering and confidence estimation yet still struggle with complex boundary and pixel topological relationships. BAPA-Net [53] introduced spatial information via boundary adaptation and prototype alignment, but its reliance on numerical mapping limits robustness and generalization due to lack of prior image sample information. In contrast, the proposed GNCE module introduces spatial information through similarity gradient consistency estimation and strengthens neighborhood consistency constraints in high-confidence regions to optimize feature aggregation. This effectively enhances classification

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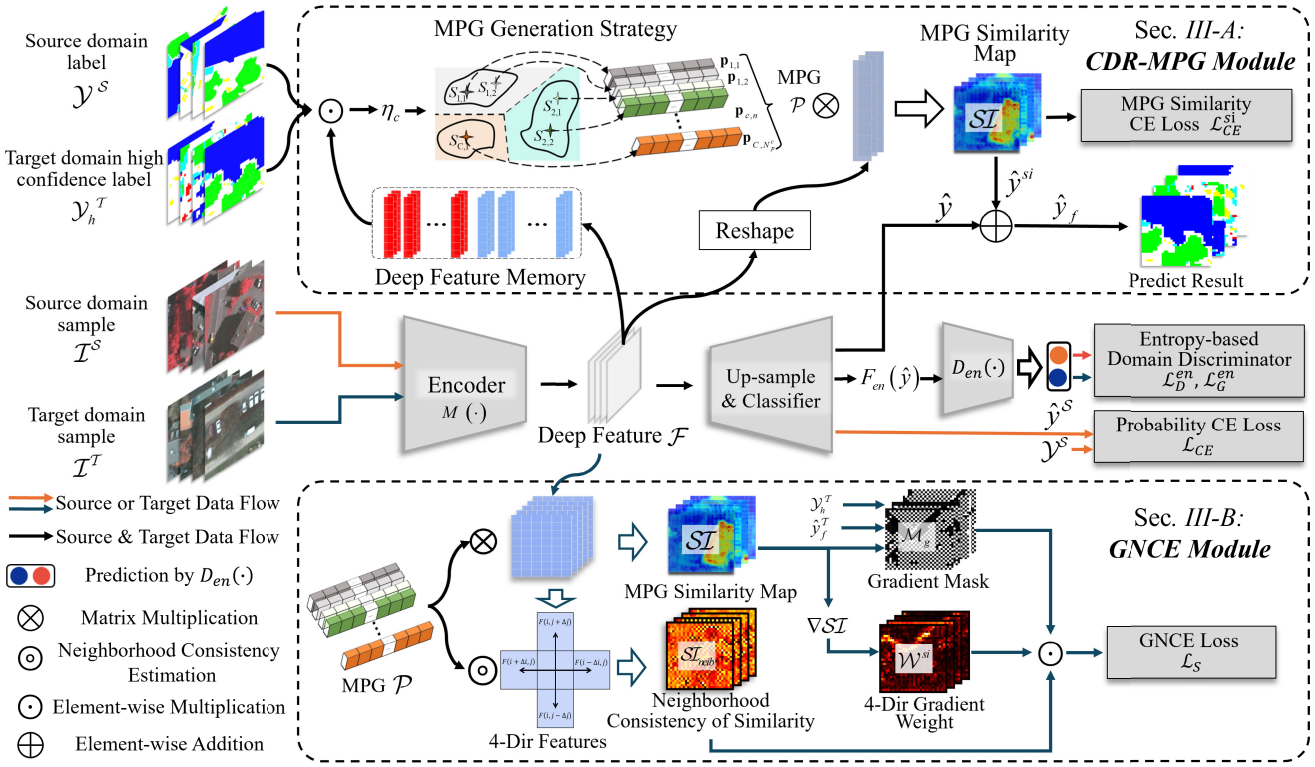


Fig. 2. Overview of the proposed NAPG model framework.

smoothness and consistency while reducing optimization interference from erroneous information.

III. Methodology

In the NAPG model, the domain $\mathcal{D}$ comprises two datasets with consistent land cover classification criteria but differing feature distributions. The source-domain dataset is defined as $\mathcal{D}_{\mathcal{S}} = (\mathcal{I}_i^{\mathcal{S}}, \mathcal{Y}_i^{\mathcal{S}})_{i=1}^{n_{\mathcal{S}}}$, where $\mathcal{I}_i^{\mathcal{S}} \in \mathbb{R}^{3\times H\times W}$ represents the ith source-domain sample, and $\mathcal{Y}_i^{\mathcal{S}} \in \mathbb{R}^{H\times W}$ denotes its corresponding label, with a total of $n_{\mathcal{S}}$ samples. The target-domain dataset is represented as $\mathcal{D}_{\mathcal{T}} = (\mathcal{I}_i^{\mathcal{T}})_{i=1}^{n_{\mathcal{T}}}$, where $\mathcal{I}_i^{\mathcal{T}}$ denotes the ith unlabeled target-domain sample, with a total of $n_{\mathcal{T}}$ samples. The goal of the NAPG model is to optimize the parameters of the RS-SS model using the proposed UDA strategy, thereby improving its performance on the target domain.

Fig. 2 illustrates the framework of the proposed NAPG model for RS-SS UDA. The NAPG model builds on the Deeplab-V3 RS-SS architecture [54], by incorporating two key UDA strategies: the CDR-MPG module and the GNCE module.

The CDR-MPG module predicts RS-SS results by calculating the similarity $\mathcal{SI}$ between the MPG $\mathcal{P}$ and the deep features $\mathcal{F}$. The MPG $\mathcal{P}$ is generated based on the deep feature sets $\mathcal{F}$ of source-domain images $\mathcal{I}^{\mathcal{S}}$ and target-domain images $\mathcal{I}^{\mathcal{T}}$, while also incorporating the corresponding source-domain labels $y^{\mathcal{S}}$ and high-confidence target-domain labels $\mathcal{I}_{lh}^{\mathcal{T}}$. The CDR-MPG module comprises the following components: 1) MPG generation strategy and 2) MPG similarity optimization function $\mathcal{L}_{\text{CE}}^{\text{si}}$. The GNCE module, as a model optimization technique, addresses the noise issue in target-domain prediction by estimating neighborhood gradient consistency. It includes the following components: 1) neighborhood consistency $\mathcal{SI}_{\text{neib}}$, which is derived from the similarity in four neighborhood directions; 2) gradient mask $\mathcal{M}_g$, which is calculated based on the gradient relationships in the similarity map and class topology information; and 3) gradient weight $\mathcal{W}^{\text{si}}$, which is generated from $\mathcal{SI}_{\text{neib}}$. Based on these components, the GNCE module is optimized through the similarity constraint loss $\mathcal{L}_{\mathcal{S}}$.

A. Cross Domain Representation of MPG

The distribution of land cover objects in remote sensing images is highly complex, and a single prototype feature is insufficient to effectively represent these objects. This limitation increases the susceptibility to noise and outliers. In the NAPG model, the MPG is a collection of prototype features representing different classes, with each class having a specified number of prototype features to align with its feature distribution. The MPG is defined as $\mathcal{P} = \{\mathcal{P}_1, \ldots, \mathcal{P}_c, \ldots, \mathcal{P}_C\}$, where the prototype group for a particular class c is $\mathcal{P}_c = \{\mathbf{p}_{c,1}, \ldots, \mathbf{p}_{c,n}, \ldots, \mathbf{p}_{c,N_p^c}\}$. Here, $\mathbf{p}_{c,n}$ represents the nth prototype feature (i.e., cluster centroid) of class c, and $\mathcal{P}_c \in \mathbb{R}^{N_p^c\times 256}$, where N_p^c is the number of prototype features for class c.

The prototype features are derived using the K-means clustering [50] based on the Euclidean distance metric applied to the normalized deep features $\tilde{\mathcal{F}}_c$ of class c. K-means minimizes the squared Euclidean distance of each point to its nearest centroid, dividing the features of class c into N_p^c

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clusters, where $c = 1, \ldots, C$. The centroid of each cluster serves as the prototype feature $\mathbf{p}_{c,n}$ for that class, computed as

$$\begin{aligned}
\mathbf{p}_{c,n} &= \frac{1}{|S_{c,n}|} \sum_{\mathcal{F}_{c,i} \in S_{c,n}} \tilde{\mathcal{F}}_{c,j} \\
\tilde{\mathcal{F}}_{c,j} &= \frac{\mathcal{F}_{c,j}}{\|\mathcal{F}_{c,j}\|_2 + \varepsilon}, \quad \tilde{\mathcal{F}}_{c,j} \in [-1, 1] \\
\mathcal{F}_{c,i} &= \{M(\mathcal{I}_i) | y_i = c\} \\
\mathcal{I}_i &\in \{\mathcal{I}^{\mathcal{S}}, \mathcal{I}^{\mathcal{T}}\} \\
y_i &\in \{\mathcal{Y}^{\mathcal{S}}, \mathcal{Y}_h^{\mathcal{T}}\}, \quad \mathcal{Y}_h^{\mathcal{T}} = \{y_{h,i}^{\mathcal{T}} | \hat{y}_i^{\mathcal{T}} \geq \tau\}
\end{aligned} \tag{1}$$

where $\mathbf{p}_{c,n} \in \mathbb{R}^{1\times 256}$ represents the nth prototype feature of class c, $S_{c,n}$ denotes the set of deep features belonging to the nth cluster of class c, $|S_{c,n}|$ is the number of elements in this cluster, and $\mathcal{F}_{c,i}$ represents the deep feature associated with class c. The encoder $M(\cdot)$ extracts deep features $\mathcal{F} = M(\mathcal{I})$ from the input image $\mathcal{I}$, where $\mathcal{F} \in \mathbb{R}^{256\times h\times w}$, and the spatial dimensions are $h = (H/16)$ and $w = (W/16)$. The class label for each deep feature $\mathcal{F}_{c,i} \in \mathbb{R}^{256}$ is obtained from source-domain labels or pseudo-labels for the target domain. Target-domain high-confidence pseudo-labels are generated by applying a probability threshold τ to the predicted probabilities $\hat{y}_i^{\mathcal{T}}$, where τ is progressively reduced from 1 to 0.9 [11], [12].

Cosine similarity effectively captures the directional relationships between features, thereby enhancing the accuracy and robustness of matching. To ensure that the prototype features $\mathbf{p}_{c,n}$ obtained through K-means clustering can be aligned with $\mathcal{F}$ using cosine similarity, the relationship between the Euclidean distance of normalized deep features $\widetilde{\mathcal{F}}$ and prototype features $\widetilde{\mathbf{p}}_{c,n}$ is established. Based on this relationship, the cosine similarity $\mathcal{SI}_c$ of objects in category c is defined as

$$\begin{aligned}
\left\| \widetilde{\mathcal{F}}_{ij} - \widetilde{\mathbf{p}}_{c,n} \right\|^2 &= \widetilde{\mathcal{F}}_{ij} \cdot \widetilde{\mathcal{F}}_{ij} - 2\widetilde{\mathbf{p}}_{c,n} \cdot \widetilde{\mathcal{F}}_{ij} + \widetilde{\mathbf{p}}_{c,n} \cdot \widetilde{\mathbf{p}}_{c,n} \\
&= 1 - 2\mathcal{SI}_{ij,c} + 1 = 2\left(1 - \mathcal{SI}_{ij,c}\right)
\end{aligned} \tag{2}$$

$$\begin{aligned}
\mathcal{SI}_{ij,c} &= \sum_{c=1}^{C} \widetilde{\mathbf{p}}_{c,n} \cdot \widetilde{\mathcal{F}}_{ij} \\
\widetilde{\mathbf{p}}_{c,n} &= \frac{\mathbf{p}_{c,n}}{\|\mathbf{p}_{c,n}\|_2 + \varepsilon}, \widetilde{\mathbf{p}}_{c,n} \in [-1, 1]
\end{aligned} \tag{3}$$

where $\varepsilon = 10^{-5}$ is used to prevent division by zero. The normalized deep feature vector at each position (i, j) is denoted as $\widetilde{\mathcal{F}}_{ij} \in \mathbb{R}^{256\times 1}$. As the cosine similarity $\mathcal{SI}_{ij,c}$ increases, the Euclidean distance $\| \widetilde{\mathcal{F}}_{ij} - \widetilde{\mathbf{p}}_{c,n} \|^2$ decreases linearly. This negative correlation indicates that the clustering effect achieved by minimizing the Euclidean distance during the K-means clustering process can be interpreted through cosine similarity. Based on the above analysis, cosine similarity is used as the matching criterion between the prototype $\mathcal{P}_c$ and the deep features $\mathcal{F}$.

Typically, as shown in Fig. 3(a), a single prototype feature is insufficient to represent land cover objects with complex intraclass feature distributions [55], [56], making it prone to mismatches with deep features. To address this limitation, the relationship between the fitting error E_c and the number of prototype features N_p^c in the prototype set $\mathcal{P}_c$ is derived based on the K-means algorithm as follows:

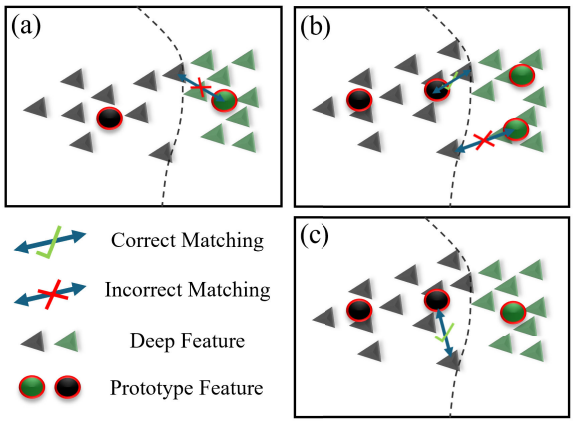


Fig. 3. Illustration of prototype and deep feature matching results. (a) Matching results with a single prototype feature. (b) Matching results with multiple prototype features of the same number across different land-cover classifications. (c) Matching results with multiple prototype features of varying numbers across different land-cover classifications.

$$E_c\left(N_p^c\right) \propto \frac{1}{N} \sum_{i=1}^{N} \min_{n\in\{1,\ldots,N_p^c\}} \left\| \widetilde{\mathcal{F}}_{ij} - \widetilde{\mathbf{p}}_{c,n} \right\|^2 \tag{4}$$

where, in the case of land cover category c, the number of pixels for deep features is N, and the ith feature to be matched is $\mathcal{F}_{c,i}$. For the land cover class c, the total number of pixels in the deep features $\mathcal{F}$ is N, and the ith feature to be matched is $\mathcal{F}_{c,i}$. For classes with highly complex feature distributions, increasing the number of prototype features N_p^c can reduce the fitting error for class c. However, for classes with less dispersed feature distributions, where features are more concentrated, using too many prototype features can cause overlap and redundancy among prototypes. As shown in Fig. 3(b), this redundancy can cause prototype features to mistakenly match deep features from other classes, increasing the mismatch rate. Therefore, the number of prototype features N_p^c should be carefully set for each class, based on the dispersion of the deep feature distribution, to effectively balance the fitting error and mismatch rate.

Since a prototype feature $\mathbf{p}_{c,n}$ in the MPG approximates the mean feature $\mu_{c,n}$ of the nth cluster, i.e., $\mathbf{p}_{c,n} \approx \mu_{c,n}$, the variance δ_c^2 can be regarded as the mean of the squared Euclidean distances from the samples to the prototype feature

$$\delta_c^2 = \frac{1}{n_c} \sum_{i=1}^{N_c} \left\| \mathcal{F}_{c,i} - \mu_{c,n} \right\|^2 \approx \frac{1}{n_c} \sum_{i=1}^{N_c} \left\| \mathcal{F}_{c,i} - \mathbf{p}_{c,n} \right\|^2. \tag{5}$$

As shown in Fig. 3(c), assigning different numbers of prototype features to different land cover classes based on their feature dispersion is an effective strategy. To quantify the dispersion of the deep feature distribution $\mathcal{F}_c$ for each land cover class, the variance δ_c^2 is calculated. These variance values δ_c^2 are then normalized to proportions, which are used to determine the required number of prototype features for each class. Variance is chosen as the metric for dispersion because it directly measures the average squared deviation from the

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mean, providing a robust and computationally efficient estimate of feature spread. The normalized variance proportion η_c is calculated using the following formula:

$$\eta_c = \left\lfloor \frac{\delta_c^2}{\sum_{i=1}^{C} \delta_i^2} \right\rfloor \tag{6}$$

where C is the total number of classes. Based on η_c, the number of prototype features for each class is determined by taking the floor $\lfloor \cdot \rfloor$ of η_c. For example, if the variance proportions of the deep features are [1.0, 1.1, 1.4, 2.4, 4.1, 4.2], the number of prototype features for each class, after retaining the integer part, is set to [1, 1, 1, 2, 4, 4]. To ensure balanced training of prototype features across classes, the prototype counts within the group are adjusted to their least common multiple by duplicating prototype features. For example, the configuration [1, 1, 1, 2, 4, 4] is expanded to [4, 4, 4, 4, 4, 4].

Traditionally, Euclidean distance is often used to optimize the alignment between prototypes and deep features, but it is sensitive to feature scaling, unable to capture the directional relationships between features, and performs poorly in high-dimensional spaces [57]. In contrast, the cross-entropy (CE) function provides more stable optimization of classification boundaries and enhances the model's generalization ability [58]. By processing the similarity graph with the SoftMax function to generate similarity probabilities and using the similarity cross-entropy CE loss $\mathcal{L}_{\mathrm{CE}}^{\mathrm{si}}$ to measure the difference between the predicted similarity probabilities $\hat{y}^{\mathrm{si}}$ and the source domain labels $\mathcal{I}_l^{\mathcal{S}}$

$$\begin{aligned} \mathcal{L}_{\mathrm{CE}}^{\mathrm{si}} &= -\sum_{i,j}\sum_{c} \mathcal{I}_{l,i,j,c}^{\mathcal{S}} \log\left(\hat{y}_{i,j,c}^{\mathrm{si}}\right) \\ \hat{y}_c^{\mathrm{si}} &= \frac{\exp(\mathcal{S}\mathcal{I}_c)}{\sum_c \exp(\mathcal{S}\mathcal{I}_c)} \end{aligned} \tag{7}$$

where $\hat{y}_c^{\mathrm{si}}$ represents the probability map obtained by applying the SoftMax activation function to the similarity map $\mathcal{S}\mathcal{I}_c$. Appendix A provides the detailed derivations of the prototype Euclidean distance and similarity CE loss optimizations, highlighting the advantages of using similarity CE. Since the number of prototype features varies across classes, the prototype feature with the highest similarity probability is selected for optimization. The corresponding formula is

$$\mathcal{L}_{\mathrm{CE}}^{\mathrm{si}} = -\sum_{i,j}\sum_{c} y_{i,j,c} \log\left(\max\left(\hat{y}_{i,j,c}^{\mathrm{si}}, \ldots, \hat{y}_{i,j,c+N_p^c}^{\mathrm{si}}\right)\right). \tag{8}$$

The CE loss $\mathcal{L}_{\mathrm{CE}}$ and the entropy-based domain discriminator $D_{\mathrm{en}}(\cdot)$, widely regarded as effective optimization techniques in the field of UDA [10], [13], [14], [15], [31], [32], [33], [59], [60], are integrated into the proposed NAPG model. The CE loss $\mathcal{L}_{\mathrm{CE}}$ for the source domain is defined as the loss between the prediction probability $\hat{y}^{\mathcal{S}}$, generated by the classifier and the corresponding source domain label $y^{\mathcal{S}}$

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{i,j}\sum_{c} y_{i,j,c}^{\mathcal{S}} \log\left(\hat{y}_{i,j,c}^{\mathcal{S}}\right). \tag{9}$$

$D_{\mathrm{en}}(\cdot)$ achieves implicit marginal distribution alignment by adversarial training to make the entropy distribution of the target domain close to that of the source domain, implicitly pushing the feature distribution of the target domain to align with the source domain. The optimization function of $D_{\mathrm{en}}(\cdot)$ is defined as

$$\begin{aligned} \mathcal{L}_D^{\mathrm{en}} = &-\left[\log\left(D_{\mathrm{en}}\left(F_{\mathrm{en}}\left(\hat{y}^{\mathcal{S}}\right)\right)\right)\right] \\ &-\left[\log\left(1 - D_{\mathrm{en}}\left(F_{\mathrm{en}}\left(\hat{y}^{\mathcal{T}}\right)\right)\right)\right] \\ F_{\mathrm{en}}\left(\hat{y}\right) = &-\sum_{c=1}^{C} y \log_2\left(\hat{y}\right), 0 \le F_{\mathrm{en}}\left(\hat{y}\right) \le 1 \end{aligned} \tag{10}$$

where $F_{\mathrm{en}}(\hat{p})$ represents the entropy of the predicted probability $\hat{y}$, measuring the uncertainty of the predicted probability distribution. The predicted probability distributions $\hat{y}^{\mathcal{S}}$ and $\hat{y}^{\mathcal{T}}$ are the outputs of the classifier for the source domain and the target domain, respectively. When the parameters of $D_{\mathrm{en}}(\cdot)$ are frozen, the optimization function is adjusted as follows:

$$\mathcal{L}_G^{\mathrm{en}} = -\mu \mathbb{E}\left[\log\left(D_{\mathrm{en}}\left(F_{\mathrm{en}}\left(\hat{y}^{\mathcal{T}}\right)\right)\right)\right] \tag{11}$$

where μ is a weighting parameter, commonly set as $\mu = 0.001$ [15], [32], [40].

Finally, to integrate the prediction results of the DeepLab-V3 classifier, which encodes spatial information, with the results of prototype similarity matching, the target-domain prediction $\hat{y}_f$ is computed as the weighted sum of the probability predictions $\hat{y}$ and the prototype similarity prediction probability $\hat{y}^{\mathrm{si}}$. The formula is expressed as follows:

$$\hat{y}_f = \arg\max\left(N_c^{\mathrm{opt}} \cdot \hat{y}^{\mathrm{si}} + \hat{y}\right) \tag{12}$$

where the optimal prototype feature number N_c^{opt} is determined by adjusting the prototype feature counts of each category to their least common multiple, thereby achieving a balanced quantity that ensures fair representation of different categories during training. For instance, when the MPG counts are adjusted to [4, 4, 4, 4, 4, 4], the optimal prototype feature number is $N_c^{\mathrm{opt}} = 4$.

B. Gradient Neighborhood Consistency Estimation

MPG serves as representative features for various land cover classes, which can better adapt to the sample distribution in the unlabeled target domain, thereby enhancing the performance of UDA. However, as shown in Fig. 1, since clustering is performed based on categories, the generated prototype features mainly represent category information, resulting in the loss of spatial information. This in turn leads to isolated misclassification noise points in the classification output. The reasons are as follows: 1) the limitations of similarity measurement. When classifying based solely on the similarity between pixels and prototype features, it is easy to be interfered with, causing pixels of different classes to be misclassified and generating misclassification noise; and 2) the topological relationships and internal distributions of land cover classes in remote sensing images are complex, and the features are discrete. Prototype features have difficulty capturing the spatial distribution patterns of samples, making isolated pixels prone to misclassification and forming isolated misclassification noise points.

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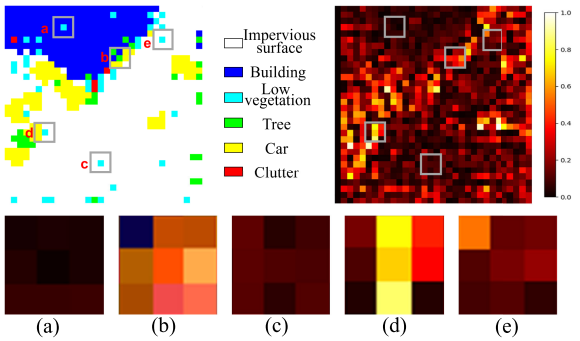


Fig. 4. Gradient map of prototype similarity corresponding to the predicted results in Fig. 1. The subplots from "a" to "b" in the second row represent the prototype similarity gradient maps within a 3 × 3 range centered on the misclassified points.

Specifically, in prototype-based RS-SS methods, isolated misclassified points usually have similar features to their neighborhoods. Especially in the prototype similarity map $\mathcal{SI}$, the neighborhoods of isolated misclassified points typically show little change in gradient. As shown in Fig. 4, which displays the classes and their corresponding gradient maps originating from Fig. 1, the discrete misclassified points "a" and "c" located within larger land cover areas, such as "Impe. Sur." and "Buil.," objects show lower gradient values in their neighborhoods. In contrast, the boundary points "b," "d," and "e" located at the boundaries of different classes show relatively higher neighborhood gradient values.

Based on the aforementioned motivations, the GNCE module is proposed. The GNCE module is designed to address the issue of isolated misclassification noise points in prototype-based RS-SS methods. By integrating similarity gradient consistency estimation with spatial information, this module enhances the aggregation of neighborhood features with their corresponding prototypes, thereby alleviating the problem of intraclass discrete misalignment. The GNCE module calculates neighborhood similarity and employs similarity gradient-based weights to capture local similarity changes, thereby improving the smoothness and consistency of the classification results. Additionally, a gradient mask is introduced to dynamically adjust the loss function, ensuring that neighborhood consistency constraints are applied only under high-confidence conditions, thereby mitigating the interference of low-confidence land cover topological boundary samples on model optimization.

For the deep feature $\mathcal{F} \in \mathbb{R}^{128\times h\times w}$, where the feature of a certain point is denoted as $\mathcal{F}(i, j)$, the neighborhood elements within the range $(\Delta i, \Delta j)$ are represented as $\mathcal{F}(i+\Delta i, j+\Delta j)$. According to (2), the neighborhood similarity $\mathcal{SI}_{\text{neib}}$ between the feature point $\mathcal{F}(i+\Delta i, j+\Delta j)$ and the most similar prototype feature $\mathbf{p}_{c,n'}$ of category c can be calculated, which is specifically expressed as follows:

$$\begin{aligned}\mathcal{SI}_{\text{neib},(i,j)} &= \widetilde{\mathbf{p}}_{c,n'} \cdot \widetilde{\mathcal{F}}(i+\Delta i, j+\Delta j)\\ n' &= \underset{n=1,\ldots,N_p^c}{\arg\max}\left(\widetilde{\mathbf{p}}_{c,n} \cdot \widetilde{\mathcal{F}}(i, j)\right)\end{aligned} \tag{13}$$

where $\mathcal{SI}_{\text{neib}} \in \mathbb{R}^{(4\times C)\times h\times w}$ represents the neighborhood similarity. The values of Δi and Δj are determined by the pixel regions of the target classes in the target domain. As an unsupervised training process, $\mathcal{SI}_{\text{neib}}$ is directly influenced by the spatial topological relationship between the center point and the neighborhood points. Therefore, a neighborhood consistency weight based on similarity gradient, $\mathcal{W}^{\text{si}}$, is proposed to measure the degree of local similarity feature changes. The similarity gradient not only reflects the local changes of features but also takes into account the relative changes between features and their corresponding prototype features. By using the similarity gradient as a weight, the similarity changes between features within the neighborhood are better captured, thereby improving the smoothness and consistency of the classification results. The gradient of the similarity map $\mathcal{SI}$ at point (i, j), $\nabla\mathcal{SI}_{(i,j)}$, is defined as follows:

$$\begin{aligned}\nabla\mathcal{SI}_{(i,j)} &= \left(\frac{\partial\mathcal{SI}}{\partial x}, \frac{\partial\mathcal{SI}}{\partial y}\right)_{(i,j)}\\ &= \frac{\partial}{\partial x}\left(\frac{\mathbf{p}_{c,n}\cdot\mathcal{F}(i,j)}{\|\mathbf{p}_{c,n}\|\|\mathcal{F}(i,j)\|}\right)_{(i,j)}\\ &\quad + \frac{\partial}{\partial y}\left(\frac{\mathbf{p}_{c,n}\cdot\mathcal{F}(i,j)}{\|\mathbf{p}_{c,n}\|\|\mathcal{F}(i,j)\|}\right)_{(i,j)}.\end{aligned} \tag{14}$$

The magnitude of the gradient reflects the variation in similarity at that point and its neighborhood. To capture the directional information of the gradient, the gradient components in the X- and Y-directions at the center point are calculated as follows:

$$\begin{aligned}\nabla\mathcal{SI}_{Y,(i,j)} &= \left|\mathcal{SI}_{(i+\Delta i,j)} - \mathcal{SI}_{(i-\Delta i,j)}\right|,\\ \nabla\mathcal{SI}_{X,(i,j)} &= \left|\mathcal{SI}_{(i,j+\Delta j)} - \mathcal{SI}_{(i,j-\Delta j)}\right|.\end{aligned} \tag{15}$$

The gradient-based weight $w_{(i,j)}$ is defined as follows:

$$\begin{aligned}w_{u,(i,j)} &= \exp\left(-\left\|\nabla\mathcal{SI}_{y,(i-\Delta i,j)}\right\|\right),\\ w_{d,(i,j)} &= \exp\left(-\left\|\nabla\mathcal{SI}_{y,(i+\Delta i,j)}\right\|\right),\\ w_{l,(i,j)} &= \exp\left(-\left\|\nabla\mathcal{SI}_{x,(i,j-\Delta j)}\right\|\right),\\ w_{r,(i,j)} &= \exp\left(-\left\|\nabla\mathcal{SI}_{x,(i,j+\Delta j)}\right\|\right).\end{aligned} \tag{16}$$

The four gradient directions (up, down, left, and right) are denoted by $\mathcal{W}_{\text{si},(i,j)} = \{w_{u,(i,j)}, w_{d,(i,j)}, w_{l,(i,j)}, w_{r,(i,j)}\}$, with $\mathcal{W}_{\text{si}} \in \mathbb{R}^{(4\times C)\times h\times w}$. The exponential function $\exp(\cdot)$ is employed to reduce the weight of points with large gradients (i.e., points with significant similarity variation), while increasing the weight of points with small gradients (i.e., points with smooth similarity variation).

To ensure that the neighborhood consistency constraint is only imposed under high-confidence conditions, thereby alleviating the interference of low-confidence land cover topological boundary samples on model optimization, a gradient mask $\mathcal{M}_g$ is introduced to dynamically adjust the loss function. Specifically, the pseudo-label $\hat{y}_f \in \mathbb{R}^{1\times h\times w}$ represents the classification result at each location (i, j), denoted as $c = \hat{y}_f(i, j)$. When the gradient value corresponding to the pseudo-label at a given location satisfies $\nabla\mathcal{SI}_{c=\hat{y}_f(i,j)} \le \nabla\mathcal{SI}_{c\neq\hat{y}_f(i,j)}$, the gradient mask $\mathcal{M}_g(i, j)$ in the corresponding direction is assigned a value of 1; otherwise, it is assigned a value of 0. The four channels of $\mathcal{M}_g \in \mathbb{R}^{4\times h\times w}$ represent the

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gradient decision results in four directions. The formula for calculating $\mathcal{M}_g$ is

$$\mathcal{M}_{g,(i,j)} = \begin{cases} 1, & \nabla \mathcal{SI}_{c=\hat{y}_f(i,j)} \leq \max_{c' \neq \hat{y}_f(i,j)} (\nabla \mathcal{SI}_{c'}) \\ 0, & \text{Otherwise.} \end{cases} \tag{17}$$

In summary, the GNCE loss is defined by

$$\mathcal{L}_S = \frac{1}{\sum_{i,j} \mathcal{M}_{g,(i,j)}} \sum_{i,j} \mathcal{M}_{g,(i,j)} \left(1 - \sum_{c}^{C} \left(\mathcal{W}_{\text{si},(i,j)} \mathcal{SI}_{\text{neib},(i,j)}\right)\right). \tag{18}$$

C. Optimization Strategy

The optimization strategy and prototype feature generation process of the proposed NAPG model are as follows: first, adaptive training of prototype features is realized by using the supervision optimization training of the source domain and the entropy discriminator. In this stage, the NAPG model integrates single-prototype features from the source domain, as shown in the feature distribution diagram in Fig. 5(a). Then, the MPG with stronger representation ability is generated by using the dispersion degree of the feature distribution of different classes in the source domain and high-confidence target-domain samples, as shown in Fig. 5(b). Finally, based on the proposed UDA strategy, the multiprototype feature set is further optimized with the evolution of the high-confidence target-domain feature distribution to achieve better alignment and adaptability, as shown in Fig. 5(c).

The training process of the NAPG model consists of two UDA training stages, with the algorithm flow illustrated in Algorithm 1. The training process includes the following.

1) *Warm-up training with single-prototype group representation*: In this stage, supervised training is performed using the source-domain data and single-prototype group, while marginal distribution alignment is achieved by an entropy-based domain adversarial module. This process not only enables the model to perform RS-SS based on prototype similarity but also preliminarily improves its adaptability to the target domain.
2) *Generation of MPG and GNCE optimization*: Based on the deep feature distributions of the source domain and the high-confidence target domain from the first stage, different numbers of MPG for different land cover classes are generated and optimized. At the same time, the proposed neighborhood consistency estimation is used to improve the optimization by complementing the spatial topological information of the prototype features.

Algorithm 1 Optimization Strategy for NAPG Model

Input: labeled source data $(\mathcal{I}^{\mathcal{S}}, \mathcal{Y}^{\mathcal{S}})$, $y^{\mathcal{S}} \in \mathbb{R}^{W \times H}$; unlabeled target data $(\mathcal{I}^{\mathcal{T}}, \emptyset)$, $\mathcal{I}^{\mathcal{T}} \in \mathbb{R}^{3 \times W \times H}$; total epoch number $EpoT$, number of source domain samples $N_{\mathcal{S}}$.
Output: UDA optimized RS-SS model $\mathcal{M}(\cdot)$.

Warm-Up training
Initialize discriminator parameters $D_{\text{en}}(\cdot)$ randomly using a standard normal distribution. Load pretrained model parameters for backbone model $M(\cdot)$, which is pretrained on the ImageNet dataset [61].
for each training epoch $n = 0$ to $\frac{EpoT}{2}$ **do**
  for each inference iteration $t = 0$ to $N_{\mathcal{S}}$ **do**
    $\mathcal{I}_t^{\mathcal{S}}, y_t^{\mathcal{S}} = fetch(\mathcal{I}^{\mathcal{S}}, \mathcal{Y}^{\mathcal{S}})$
    $\mathcal{F}_t^{\mathcal{S}} \leftarrow \mathcal{M}(\mathcal{I}_t^{\mathcal{S}})$ #Deep feature
  end for
  $\mathcal{P} \leftarrow KMeans(\mathcal{F}^{\mathcal{S}}, \mathcal{Y}^{\mathcal{S}})$ #Single prototype group
  for each inference iteration $t = 0$ to $N_{\mathcal{S}}$ **do**
    $\mathcal{I}_t^{\mathcal{S}}, y_t^{\mathcal{S}} \leftarrow fetch(\mathcal{I}^{\mathcal{S}}, \mathcal{Y}^{\mathcal{S}})$, $\mathcal{I}_t^{\mathcal{T}} \leftarrow fetch(\mathcal{I}^{\mathcal{T}})$
    $\hat{y}_t^{\text{si},\mathcal{S}}, \hat{y}_t^{\mathcal{S}} \leftarrow \mathcal{M}(\mathcal{I}_t^{\mathcal{S}}, \mathcal{P})$, $\hat{y}_t^{\text{si},\mathcal{T}}, \hat{y}_t^{\mathcal{T}} \leftarrow \mathcal{M}(\mathcal{I}_t^{\mathcal{T}}, \mathcal{P})$#Eq. (7)
    $\mathcal{L}_{CE,t}^{\text{si},\mathcal{S}} = \mathcal{L}_{CE}^{\text{si}}(\hat{y}_t^{\text{si},\mathcal{S}}, y_t^{\mathcal{S}})$ #Eq. (3), (8)
    $\mathcal{L}_{CE,t}^{\mathcal{S}} = \mathcal{L}_{CE}(\hat{y}_t^{\mathcal{S}}, y_t^{\mathcal{S}})$#Eq. 9
    $\mathcal{L}_{D,t}^{\text{en}} = \mathcal{L}_D^{\text{en}}(D_{\text{en}}(\hat{y}_t^{\mathcal{S}}), D_{\text{en}}(\hat{y}_t^{\mathcal{T}}))$ #Eq. (10)
    $\mathcal{L}_{G,t}^{\text{en}} = \mathcal{L}_G^{\text{en}}(D_{\text{en}}(\hat{y}_t^{\mathcal{T}}))$ # Frozen $D_{\text{en}}(\cdot)$.Eq.(11)
    Update $\mathcal{M}(\cdot)$ and $D_{\text{en}}(\cdot)$.
  end for
end for
Generation of MPG and GNCE optimization
for each training epoch $n = \frac{EpoT}{2}$ to $EpoT$ **do**
  for each inference iteration $t = 0$ to $N_{\mathcal{S}}$ **do**
    $\mathcal{I}_t^{\mathcal{S}}, y_t^{\mathcal{S}} \leftarrow fetch(\mathcal{I}^{\mathcal{S}}, \mathcal{Y}^{\mathcal{S}})$
    $\mathcal{I}_t^{\mathcal{T}} \leftarrow fetch(\mathcal{I}^{\mathcal{T}})$
    $\mathcal{F}_t^{\mathcal{S}} \leftarrow \mathcal{M}(\mathcal{I}_t^{\mathcal{S}})$, $F_t^{\mathcal{T}}, \hat{y}_t^{\mathcal{T}} \leftarrow M(\mathcal{I}_t^{\mathcal{T}})$
    $\mathcal{Y}_{lh,t}^{\mathcal{T}} \leftarrow PseudoLabelGenerator(\hat{y}_t^{\mathcal{T}}, \tau)$ #Eq.(1)
  end for
  # update MPG
  $\eta_c \leftarrow Var(\{\mathcal{F}^{\mathcal{S}}, \mathcal{F}^{\mathcal{T}}\}, \{\mathcal{Y}^{\mathcal{S}}, \mathcal{Y}_{lh}^{\mathcal{T}}\})$#Eq. (5), (6)
  $\mathcal{P} \leftarrow KMeans(\{\mathcal{F}^{\mathcal{S}}, \mathcal{F}^{\mathcal{T}}\}, \{\mathcal{Y}^{\mathcal{S}}, \mathcal{Y}_{lh}^{\mathcal{T}}\})$
  for each inference iteration $t = 0$ to $N_{\mathcal{S}}$ **do**
    $\mathcal{I}_t^{\mathcal{S}}, y_t^{\mathcal{S}} \leftarrow fetch(\mathcal{I}^{\mathcal{S}}, \mathcal{Y}^{\mathcal{S}})$, $\mathcal{I}_t^{\mathcal{T}} \leftarrow fetch(\mathcal{I}^{\mathcal{T}})$
    $\hat{y}_t^{\text{si},\mathcal{S}}, \hat{y}_t^{\mathcal{S}}, \mathcal{SI}^{\mathcal{S}} \leftarrow M(\mathcal{I}_t^{\mathcal{S}}, \mathcal{P})$ #Eq.(7)
    $\hat{y}_t^{\text{si},\mathcal{T}}, \hat{y}_t^{\mathcal{T}}, \mathcal{SI}^{\mathcal{T}} \leftarrow M(\mathcal{I}_t^{\mathcal{T}}, \mathcal{P})$ #Eq.(7)
    $\mathcal{Y}_{lh,t}^{\mathcal{T}} \leftarrow PseudoLabelGenerator(\hat{y}_t^{\mathcal{T}}, \tau)$#Eq.(1)
    $\mathcal{L}_{CE,t}^{\text{si},\mathcal{S}} = \mathcal{L}_{CE}^{\text{si}}(\hat{y}_t^{\text{si},\mathcal{S}}, y_t^{\mathcal{S}})$ #Eq. (3), (8)
    $\mathcal{L}_{CE,t}^{\mathcal{S}} = \mathcal{L}_{CE}(\hat{y}_t^{\mathcal{S}}, y_t^{\mathcal{S}})$#Eq. (9)
    $\mathcal{L}_{D,t}^{\text{en}} = \mathcal{L}_D^{\text{en}}(D_{\text{en}}(\hat{y}_t^{\mathcal{S}}), D_{\text{en}}(\hat{y}_t^{\mathcal{T}}))$ #Eq. 10
    $\mathcal{L}_{G,t}^{\text{en}} = \mathcal{L}_G^{\text{en}}(D_{\text{en}}(\hat{y}_t^{\mathcal{T}}))$ # Eq.(11)
    # GNCE optimization
    $\mathcal{SI}_{neib,t}^{\mathcal{T}} = NeiborSim(\mathcal{F}_t^{\mathcal{T}}, \mathcal{P})$ #Eq. (13)
    $\nabla \mathcal{SI}_t^{\mathcal{T}} = Grad(\mathcal{SI}_t^{\mathcal{T}})$ #Eq. (14)
    $\mathcal{W}_{\text{si}} = \{w_u, w_d, w_l, w_r\}$ #Eq. 15, 16
    $\mathcal{M}_{g,t}^{\mathcal{T}} = GradMask(\nabla \mathcal{SI}_t^{\mathcal{T}}, \mathcal{Y}_{lh,t}^{\mathcal{T}})$ #Eq. (17)
    $\mathcal{L}_{S,t}^{\mathcal{T}} = \mathcal{L}_S(\mathcal{M}_{g,t}^{\mathcal{T}}, \mathcal{W}_{\text{si},t}^{\mathcal{T}}, \mathcal{SI}_{neib,t}^{\mathcal{T}})$ #Eq. (18)
    Update $\mathcal{M}(\cdot)$ and $D_{\text{en}}(\cdot)$.
  end for
end for

IV. Experiment

A. Experimental Datasets and Setting

The datasets used in the experiment are the publicly available Vaihingen (VAI) [62] and Potsdam (POT) [63]. These two datasets were collected by different optical remote sensing devices and obtained information from different spectral bands. The VAI and POT datasets follow the same labeling

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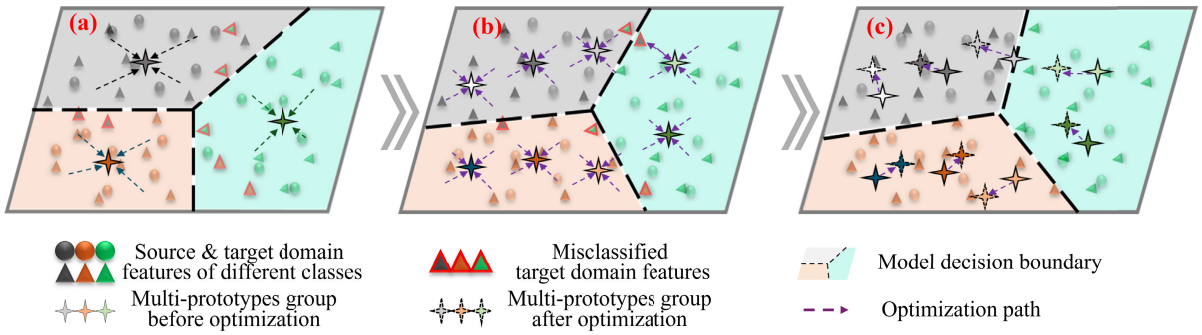


Fig. 5. Illustration of the NAPG model optimization process. (a) Warm-up training for single-prototype feature representation. (b) Generation of MPG. (c) Training of MPG reinforcement representation with GNCE optimization.

TABLE I
OVERVIEW INFORMATION OF EXPERIMENTAL DATASETS

Name	Band	Resolution	Size	Number
VAI	IR-R-G	0.09 m/pixel	2500×2000	33
POT-IRRG	IR-R-G	0.05 m/pixel	6000×6000	38
POT-RGB	R-G-B	0.05 m/pixel	6000×6000	38
LoveDA-Urban	R-G-B	0.3 m/pixel	1024×1024	2993
LoveDA-Rural	R-G-B	0.3 m/pixel	1024×1024	2994

*Note: IR: Infrared band. R: Red band. G: Green band. B: Blue band.

rules, involving six types of land cover classes, namely, impervious surface (Impe. Sur.), building (Buil.), low vegetation (Low Veg.), tree, car, and clutter. An overview of the remote sensing datasets is shown in Table I.

Based on the above datasets, four experimental dataset configurations (source domain → target domain) were constructed to validate the UDA performance of the proposed NAPG model.

1) *POT-RGB→VAI:* The source domain consists of the RGB bands from the POT dataset, while the target domain is the VAI dataset. This task involves two domains with different imaging locations, spatial resolutions, and band compositions.
2) *POT-IRRG→VAI*: The source domain consists of the IRRG bands from the POT dataset, with the VAI dataset as the target domain. This task involves domains with different imaging locations and spatial resolutions but identical band compositions.
3) *VAI→POT-IRRG*: The source domain is the VAI dataset, while the target domain consists of the IRRG bands from the POT dataset. These domains share the same band compositions but differ in imaging locations and spatial resolutions.
4) *VAI→POT-RGB*: The source domain is the VAI dataset, with the target domain consisting of the RGB bands from the POT dataset. This task involves domains that differ in imaging locations, band compositions, and spatial resolutions.

Model training was conducted on an NVIDIA 3090 GPU. During training, the batch size was set to 10, and the learning rate was configured to 0.001. The evaluation metrics for the experiment included mean intersection over union (mIoU), mean $F1$ score (mF1), and mean dice coefficient (mDice) [2]. The experiment mainly consisted of the following parts.

1) *Ablation experiment of MPG*: This experiment validated the effectiveness of the proposed MPG and analyzed the impact of different scaling factors (γ) on MPG performance.
2) *Ablation experiment of GNCE module*: This experiment validated the GNCE module by examining the influence of different neighborhood ranges on the model performance. It also explored the applicability of the module to different land cover characteristics and included visualization analysis.
3) *Performance Comparison Experiment:* This experiment evaluated and analyzed the performance advantages of the NAPG model by comparing it with nine state-of-the-art UDA models. The LoveDA dataset was used for supplementary experiments, and the details and results are presented in Appendix B.

B. Cross Domain Representation of MPG Ablation Experiment

This experiment is designed to validate the effectiveness and advantages of the proposed MPG strategy. Furthermore, it examines the impact of different scaling factors (γ), treated as common multiples, to regulate the number of prototype features and evaluate their influence on performance. To achieve these goals, the following experiments are performed.

1) *Baseline performance experiment (Warm-Up)*: This phase involves evaluating the model's performance after Step 1 training, as described in Section III-C.
2) *Performance experiment without the MPG strategy (w/o MPG)*: After the warm-up phase, experiments are conducted using the same number of prototype features for different land-cover classes without applying the MPG strategy. For example, if γ is set to 3, the number of prototype features for the six land-cover classes is [3, 3, 3, 3, 3, 3].
3) *Performance experiment with the MPG strategy (w/ MPG)*: Building on the warm-up phase, this experiment applies the proposed MPG strategy for setting the number of prototypes. For example, when the normalized prototype numbers is [1, 1, 1, 2, 4, 4], multiplying

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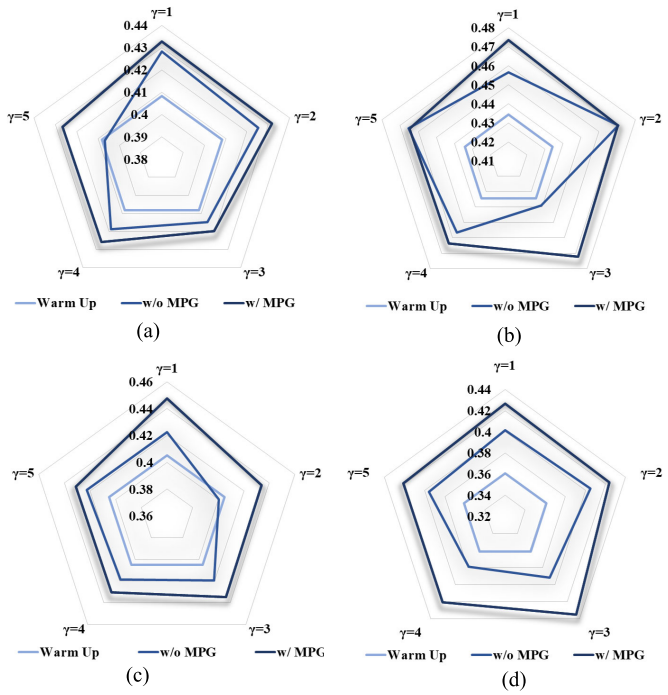


Fig. 6. Results of ablation experiments on the number and generation mode of different MPG in the four datasets. (a) mIoU of POT-RGB→VAI. (b) mIoU of POT-IRRG→VAI. (c) mIoU of VAI→POT-IRRG. (d) mIoU of VAI→POT-RGB.

by the scaling factor $\gamma = 3$ results in MPG numbers of [3, 3, 3, 6, 12, 12].

The experimental results are derived using prototype similarity matching, as shown in (7). The results of all four sets of experiments are shown in Fig. 6. The warm-up phase demonstrates that, within a limited range, it improves performance in the target domain and facilitates initial UDA in RS-SS through prototype distance matching.

In the case of "w/o MPG," the experimental results shown in Fig. 6 reveal that when the number of prototypes is the same for each land-cover class, no positive correlation is observed between the number of prototypes and UDA performance. The optimal scaling factors γ for the four datasets are $\gamma = 1$, $\gamma = 5$, $\gamma = 3$, and $\gamma = 2$, respectively. These results confirm the conclusion in Section III-A.

1) *Land-cover objects with high feature dispersion:* For objects with a high dispersion of deep features, the feature distribution is more scattered. As shown in Fig. 3(a), a single prototype feature is insufficient to represent the entire feature distribution, leading to incorrect feature matches during prototype similarity matching.
2) *Land-cover objects with low feature dispersion:* For objects with a low dispersion of deep features, the feature distribution is more concentrated. As shown in Fig. 3(b), using too many prototype features results in overlap and redundancy, increasing the likelihood of mismatches between prototype features and deep features from other classes.

Before conducting the "w/ MPG" experiments, it is necessary to calculate the proportion of prototypes η_c for different classes within the MPG based on the dispersion degree of land-cover features. After the warm-up phase, the η_c values for different classes across the four datasets [normalized and rounded as per (6)] are shown in Table II. From this table, it is evident that variance proportions are strongly related to land-cover types. Compared to classes, such as "Impe. Sur." and "Buil.," which exhibit smooth internal contours, classes, such as "car" and "clutter," show significant pixel variation, resulting in higher feature variance.

TABLE II
PROPORTION OF PROTOTYPES IN THE NORMALIZED MPG

Data	Number of prototypes
POT-RGB→VAI	1, 1, 1, 1, 4, 4
POT-IRRG→VAI	1, 1, 1, 1, 2, 2
VAI→POT-IRRG	1, 1, 1, 1, 2, 4
VAI→POT-RGB	1, 1, 3, 1, 6, 6
LoveDA-Urban → LoveDA-Rural	4, 2, 1, 1, 1, 1, 1
LoveDA-Rural → LoveDA-Urban	4, 1, 1, 1, 1, 4, 2

In the case of "w/ MPG," the experimental results shown in Fig. 6 indicate a significant performance improvement over "w/o MPG," while the variation of the scaling factor γ has a negligible impact on the UDA performance. The results show that for the three datasets, the best performance is achieved at $\gamma = 1$, and the "VAI→POT-RGB" dataset performs best at $\gamma = 3$, although with only a marginal improvement compared to $\gamma = 1$. The analysis of the experimental results suggests that simply increasing the number of MPG does not continuously enhance performance. As shown in Fig. 3(b), determining the number of MPG based on the dispersion of intraclass distributions can improve the matching accuracy between deep features and MPG, thereby enhancing the UDA capability of MPG in the target domain. Considering the balance between computational resources and performance, it is recommended to set the scaling factor to $\gamma = 1$.

C. GNCE Module Ablation Experiment

The purpose of this experiment is to validate the performance and effectiveness of the proposed GNCE module. The experiment consists of three main parts: 1) investigating the effect of different neighborhood ranges by setting different values of $(\Delta i, \Delta j)$; 2) exploring the relationship between the GNCE module and the characteristics of different ground objects; and 3) visualizing and analyzing the effectiveness of the GNCE module.

Since land-cover objects in remote sensing images often exhibit random rotational properties, the neighborhood ranges are set equally in all directions, i.e., $(\Delta i = \Delta j)$. As described in Section III-A, the deep feature dimensions are $\mathcal{F} \in \mathbb{R}^{B,128,h,w}$, and the spatial resolution of the VAI dataset is 0.09 m/pixel. Unlike convolutional networks, where the receptive field characterizes feature similarity, a single element in the deep feature map represents land-cover objects within a 1.44-m range. Given that the average length of a vehicle is 5 m, at least three pixels are required to effectively represent the "car" class. Based on this property, the neighborhood range for the same class is set as $(\Delta i = \Delta j \leq 3)$. Ablation experiments

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TABLE III
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS

Dataset	Method	mIoU(%)	mF1(%)	mDice(%)	IoU(%)						F1(%)					
					Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter	Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter
POT-RGB→VAI	D0	25.96	37.74	28.20	39.21	48.32	4.21	41.32	24.97	2.64	56.52	65.32	8.35	59.35	35.21	4.21
	D1	45.31	60.29	60.65	59.63	73.78	27.31	57.14	30.05	26.97	74.01	84.98	42.36	73.87	46.98	42.31
	D2	**49.31**	**63.50**	**63.96**	61.31	75.21	43.85	58.61	27.32	28.31	76.57	86.12	61.45	73.36	43.87	44.32
POT-IRRG→VAI	D0	31.01	44.33	44.69	36.21	52.12	16.32	55.12	32.57	6.32	53.12	68.12	27.21	71.21	35.31	11.21
	D1	49.91	63.71	64.14	62.54	80.15	47.69	62.12	28.15	20.31	77.56	89.02	64.78	76.51	43.98	33.01
	D2	**52.31**	**66.31**	**68.70**	65.32	82.32	47.35	62.32	28.35	30.35	79.36	90.12	64.32	77.35	44.31	46.31
VAI→POT-IRRG	D0	23.29	44.97	45.75	38.32	38.36	26.22	24.76	29.95	22.21	55.36	55.45	51.32	38.74	45.63	36.21
	D1	43.15	58.60	59.21	53.05	61.87	45.25	39.69	44.48	16.81	69.57	75.09	62.38	56.15	61.58	28.46
	D2	**44.72**	**59.82**	**60.39**	49.65	67.98	49.32	48.57	42.87	13.36	66.14	80.98	66.65	65.32	59.47	23.64
VAI→POT-RGB	D0	36.01	52.28	53.06	44.35	49.81	43.32	28.78	30.97	23.47	61.78	66.31	60.82	44.21	46.54	37.19
	D1	48.31	64.36	64.88	48.78	64.52	52.90	52.06	46.68	28.21	65.25	78.46	68.51	69.25	63.98	44.65
	D2	**49.32**	**63.32**	**66.05**	54.35	67.25	53.45	51.64	42.35	31.74	70.65	80.14	69.45	68.65	59.87	47.32

*Note: **Bolded data** represents the best results.

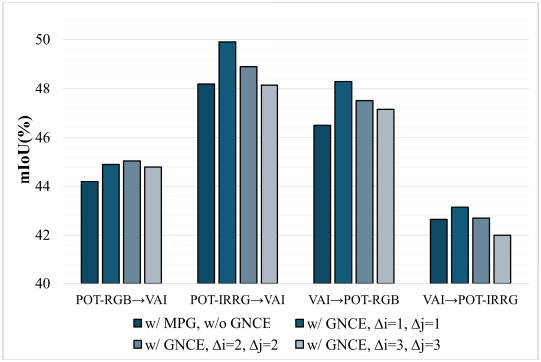


Fig. 7. Ablation study results of the GNCE module under different neighborhood ranges.

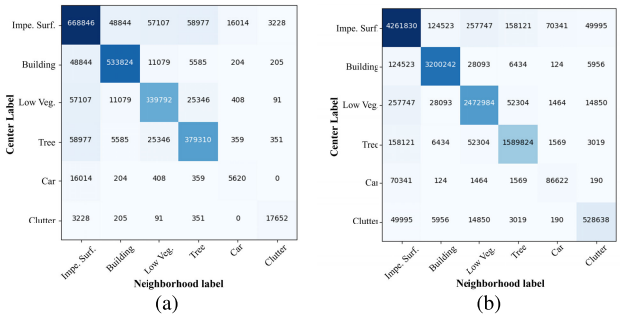


Fig. 8. Confusion matrix between neighboring labels within a pixel. (a) POT dataset. (b) VAI dataset.

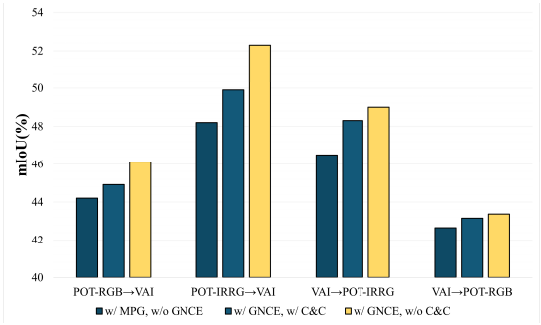


Fig. 9. Performance comparison of GNCE module with different land cover objects.

were conducted to analyze the impact of different neighborhood ranges (Δi, Δj) on model performance, with the results shown in Fig. 7. These results demonstrate that incorporating the GNCE module improves model performance, with the most significant enhancement observed at $\Delta i = \Delta j = 1$. However, performance decreases slightly as the neighborhood range increases, probably because the range directly affects the topological relationships between neighboring points. Small-scale land cover objects, such as "clutter" and "car," can interfere with neighborhood gradient optimization when the range becomes too large.

To demonstrate the performance of the different modules and their improvement effects, the detailed performance data of the ablation experiments for different modules are organized in Table III. In this table, D0 represents the performance of the target domain when trained only on the source domain data; D1 indicates the performance of the target domain after adding the MPG module; and D2 refers to the performance status after adding both the MPG and GNCE modules simultaneously.

To investigate the topological relationships between different land cover labels in the POT and VAI datasets, the class relationships between the center pixel and neighboring pixels within a 3×3 grid were analyzed. A label neighborhood confusion matrix was constructed, as shown in Fig. 8. The results show that the class "Impe. Sur." class has the largest proportion of single object areas, while the "car" class has the smallest. Additionally, due to their smaller object areas, "car" and "clutter" are less likely to appear as neighboring land cover objects.

To evaluate the influence of different land cover classes on the performance of the GNCE module, two gradient consistency experimental strategies were designed based on the above analysis to evaluate the influence of the "car" and "clutter" classes.

1) *GNCE w/ C&C*: Under the condition ($\Delta i = \Delta j = 1$), GNCE optimization is applied to all classes, including "car" and "clutter."
2) *GNCE w/o C&C*: Under the condition ($\Delta i = \Delta j = 1$), GNCE optimization excludes "car" and "clutter" classes that have weaker neighborhood topological relations and is applied only to other classes.

Using the performance of the optimal MPG ("w/ MPG") shown in Fig. 6 as a baseline, Fig. 9 shows the comparative

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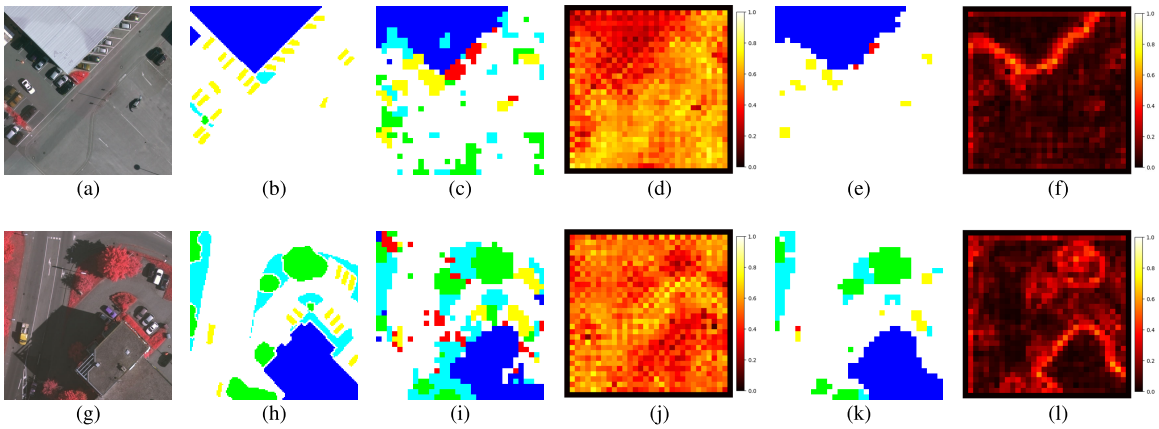


Fig. 10. Performance comparison for "POT-RGB→VAI" samples. From left to right: (a) and (g) original data, (b) and (h) present the ground truth, (c) and (i) display the predictions before optimization, (d) and (j) illustrate the normalized neighborhood gradient maps before optimization, (e) and (k) predictions after optimization, and (f) and (l) present the normalized neighborhood gradient maps after optimization.

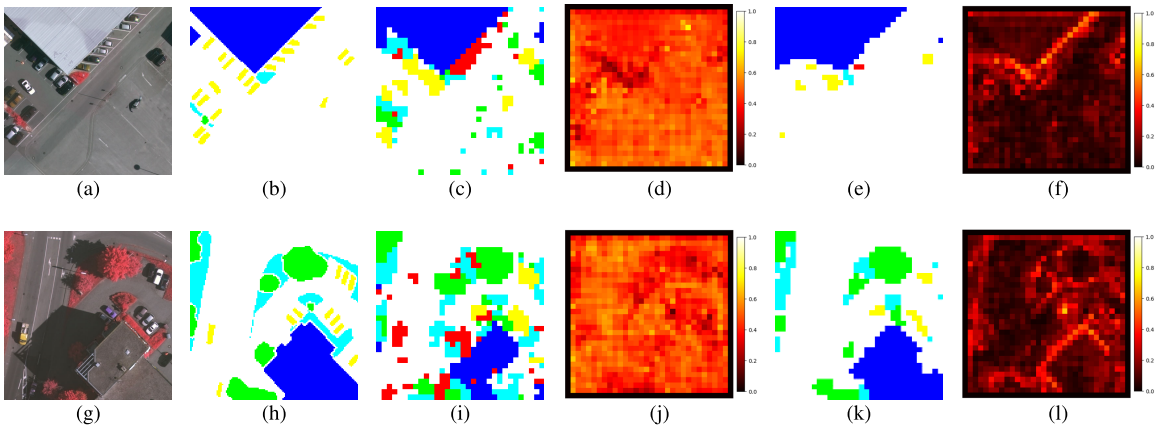


Fig. 11. Performance comparison for "POT-IRRG→VAI" samples. From left to right: (a) and (g) original data, (b) and (h) present the ground truth, (c) and (i) display the predictions before optimization, (d) and (j) illustrate the normalized neighborhood gradient maps before optimization, (e) and (k) predictions after optimization, and (f) and (l) present the normalized neighborhood gradient maps after optimization.

results of "GNCE w/o C&C" and "GNCE w/ C&C." The experimental results are summarized as follows: 1) Overall GNCE performance: the GNCE module significantly improves the UDA performance, especially for the "POT-RGB→VAI" and "POT-IRRG→VAI" datasets. As shown in Fig. 8, the VAI dataset contains a significantly higher proportion of the same-class sample pixels (diagonal elements) within neighborhoods compared to cross-class sample pixels (off-diagonal elements). This indicates that larger single object areas dominate the VAI dataset, allowing the GNCE module to perform better when processing large area land cover objects. 2) Performance of "GNCE w/o C&C": the best overall model performance is achieved in the "GNCE w/o C&C" scenario. This is due to the more complex neighborhood topology of small objects compared to large objects. Although the proposed (17) filters out some inconsistencies, small object classes, such as "car" and "clutter," still negatively affect the overall performance. By excluding these discrete small-object classes, the model focuses more effectively on learning the feature relationships of large land-cover objects, thereby improving overall RS-SS performance.

The visualization results of the consistency gradient maps are shown in Figs. 10–13. These figures sequentially show the original data, the ground truth, the predictions before optimization, the normalized neighborhood gradient maps $\nabla\mathcal{SI}_{\tilde{m}}$ before optimization, the predictions after optimization, and the normalized neighborhood gradient maps $\nabla\mathcal{SI}_{\tilde{m}}$ after optimization. The normalized neighborhood gradient maps are computed by averaging the maximum similarity gradients in four directions, as defined in (19). The results show that without the GNCE module, the predictions of the model contain many discrete mismatching points. In addition, the neighborhood gradient maps show higher values, reflecting weak gradient transitions at class boundaries. In contrast, after applying the GNCE optimization, the number of mismatched points is significantly reduced. The optimized gradient maps show smaller intraclass gradients and sharper transitions at interclass boundaries. These results demonstrate that the GNCE strategy effectively improves the model's classification performance by improving intraclass consistency and enhancing distinctions at interclass boundaries

$$\nabla\mathcal{SI}_{\tilde{m}} = \frac{1}{4}\frac{\sum_{i=1}^{4}\nabla\mathcal{SI}_m}{\max\left(\sum_{i=1}^{4}\nabla\mathcal{SI}_m\right)}$$
$$m = \underset{c=1,\ldots,C}{\arg\max}(\mathcal{SI}_c). \tag{19}$$

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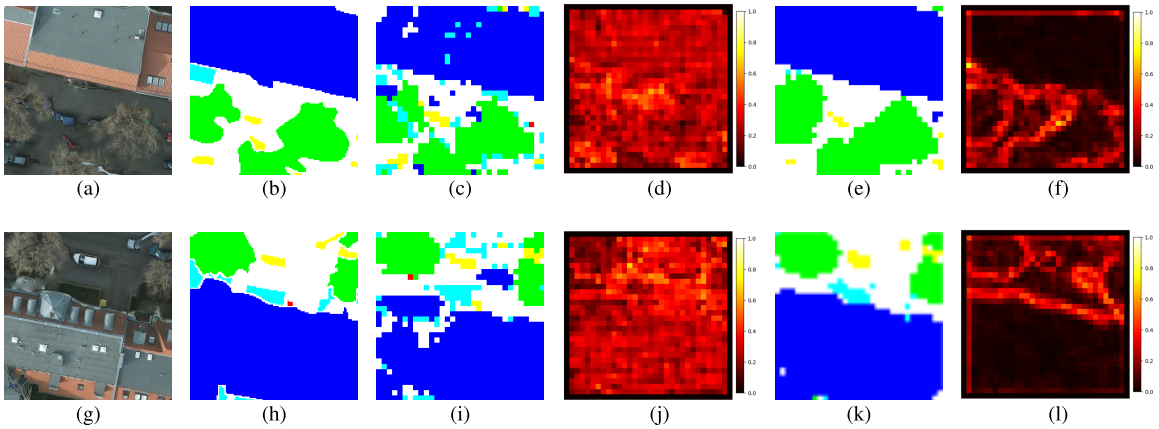


Fig. 12. Performance comparison for "VAI→POT-RGB" samples. From left to right: (a) and (g) original data, (b) and (h) present the ground truth, (c) and (i) display the predictions before optimization, (d) and (j) illustrate the normalized neighborhood gradient maps before optimization, (e) and (k) predictions after optimization, and (f) and (l) present the normalized neighborhood gradient maps after optimization.

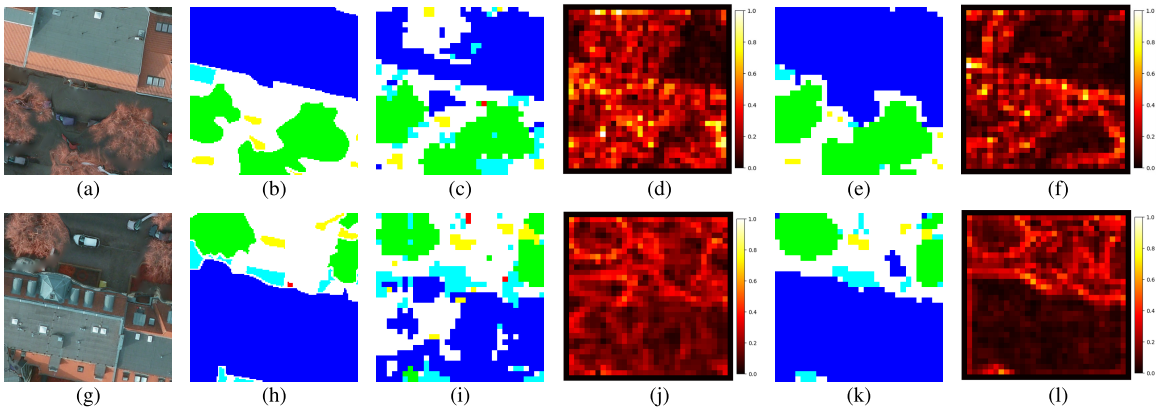


Fig. 13. Performance comparison for "VAI→POT-IRRG" samples. From left to right: (a) and (g) original data, (b) and (h) present the ground truth, (c) and (i) display the predictions before optimization, (d) and (j) illustrate the normalized neighborhood gradient maps before optimization, (e) and (k) predictions after optimization, and (f) and (l) present the normalized neighborhood gradient maps after optimization.

To further validate the optimization effects of the GNCE module on the relationships between center pixels and neighboring pixels, cosine similarity confusion matrices were computed before and after applying the GNCE module. The results for the four datasets are shown in Fig. 14. The numerical values in the confusion matrix indicate the normalized similarity values between deep features of different class labels. Without the GNCE loss $\mathcal{L}_S$, intraclass differences are relatively large (i.e., diagonal elements have lower values), and interclass differences are relatively small (i.e., off-diagonal elements have higher values). This effect is particularly noticeable between the "car" and "Impe. Sur." classes, where high deep feature similarities make these classes prone to misclassification. After introducing the GNCE module, the similarity between center pixels and neighboring pixels within the same class increases significantly (i.e., diagonal elements show higher values), while the similarity between different classes is effectively suppressed. These results further demonstrate the effectiveness of the GNCE module in enhancing intraclass consistency and reducing classification errors.

D. Performance Comparison Experiment

To validate the performance advantages of the proposed NAPG model, comparative experiments were conducted against nine state-of-the-art UDA models. The experimental results are presented in Table IV. Based on their working principles, the nine UDA algorithms were categorized into three groups: adversarial training, image translation, and prototype-based representation. For experimental consistency, all models used the ResNet-50 backbone [64], pretrained on the ImageNet dataset [61], and experiments were conducted using the DeepLab-V3 RS-SS model. As described in Section IV-A, the experiments utilized the POT and VAI datasets, defining four distinct UDA tasks. The experimental results are presented in Tables V–VIII.

MASN usually performs well in handling larger categories (such as "Buil." or "Impe. Sur.") but shows certain limitations in dealing with smaller or more complex categories (such as "clutter" or "car"). This limitation is mainly due to the insufficient capability of prototype feature representation. To overcome this limitation, CaGAN and RCA-DD integrate several UDA strategies, such as category adversarial learning, prototype representation, and attention alignment. This unified framework significantly improves model performance. DiGA combines data augmentation and adversarial learning, showing unique advantages in cross-domain transfer. This combination enables it to effectively deal with UDA challenges and provides stronger generalization ability when facing domain shifts.

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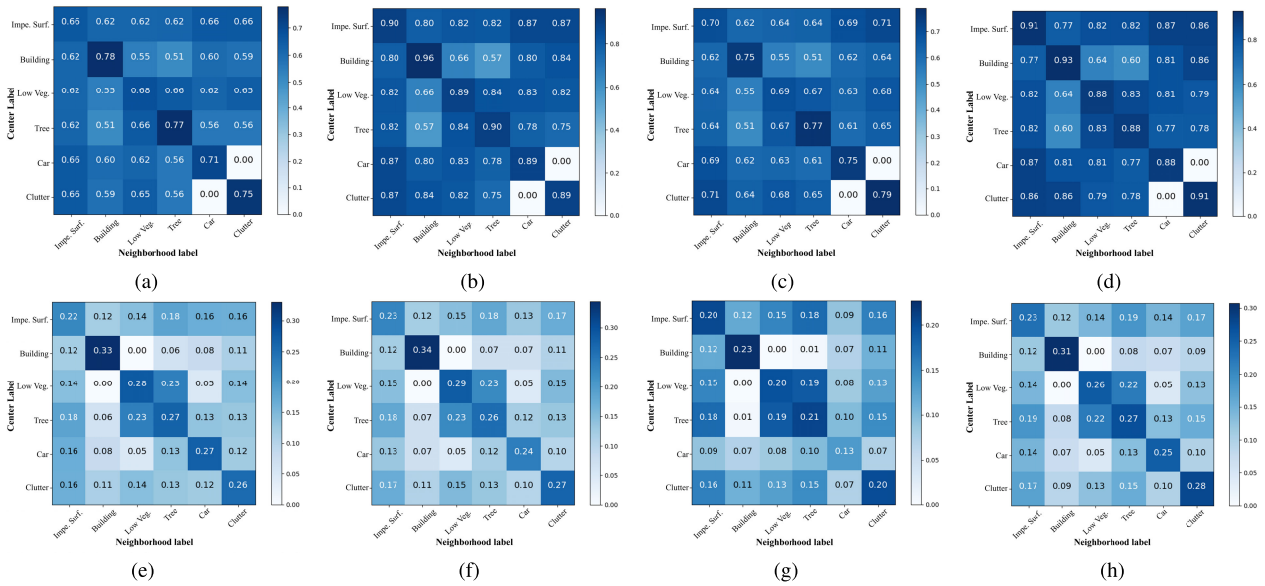


Fig. 14. Normalized similarity confusion matrix of center pixel and neighborhood pixel before and after optimization of consistency estimation. (a) POT-RGB → VAI (before). (b) POT-RGB → VAI (after). (c) POT-IRRG → VAI (before). (d) POT-IRRG → VAI (after). (e) VAI → POT-IRRG (before). (f) VAI → POT-IRRG (after). (g) VAI → POT-RGB (before). (h) VAI → POT-RGB (after).

TABLE IV

OVERVIEW OF UDA COMPARISON METHODS

Method	Source	Module	Parameters	FLOPs*	Key Words
MCD [28]	CVPR 2018	Feature Extractor, Dual Classifier	6.5×10^7	26.5G	Adversarial Learning
CLAN [29]	CVPR 2019	Feature Extractor; Classifier; Discriminator;	4.2×10^7	27.2G	class-level Adversarial Learning
CCDA [31]	TGRS 2022	Generator; Entropy & Feature Discriminator;	7.2×10^7	28.1G	Entropy & Features Adversarial Learning
CycleGAN [65]	ICCV 2017	CycleGAN	4.7×10^7	29.2G	UDA Based on Image Transfer
DiGA [9]	CVPR 2023	Cycle GAN; Data Augmentation;	6.4×10^7	32.1G	UDA Based on Image Transfer; Prototype Representation;
ProDA [39]	CVPR 2021	Feature Extractor; Classifier; Teacher-Student Distillation Moudule;	4.3×10^7	26.7G	Prototype Representation; Distillation Learning;
MASN [18]	TGRS 2023	Feature Extractor; Invariant Domain-level Memory Aggregation;	5.2×10^7	29.9G	Pseudo Label Extraction by Prototype
CaGAN [30]	TGRS 2022	Principal Component Analysis; Transferable Attention Alignment; Adaptive Class Selection & Alignment; Adaptive and Reliable class Selection;	4.6×10^7	30.5G	Prototype Representation; Class-level Assignment; Adversarial Learning
RCA-DD [40]	TGRS 2022	Domain discriminator for entropy; Regional attention module; class clearing module;	5.1×10^7	29.7G	Prototype Representation; Adversarial Learning;
Our	-	Feature Extractor; Gradient Neighborhood Consistency Estimator; Entropy Probability Cross Discriminator;	4.3×10^7	29.5G	Multi-Prototype Group Representation; Adversarial Learning; Distillation Learning

*Note: The size of the input sample is $3 \times 512 \times 512$.

TABLE V

QUANTITATIVE RESULTS OF THE "POT-RGB → VAI" SCENARIO

Methods	mIoU(%)	mF1(%)	mDice(%)	IoU(%)						F1(%)					
				Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter	Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter
Source-only	25.96	37.74	38.20	39.21	48.32	4.21	41.32	21.97	2.64	56.52	65.32	8.35	59.35	35.21	4.21
MCD	37.92	51.94	51.93	50.09	60.15	27.44	54.85	29.66	5.34	66.74	75.12	43.06	70.84	45.75	10.14
CLAN	36.34	50.10	52.16	49.00	57.63	26.81	**70.05**	27.41	3.28	65.77	73.12	42.28	69.15	43.03	6.35
CCDA	43.17	57.77	57.83	57.65	65.44	29.77	35.87	**56.99**	13.27	73.14	79.11	45.88	52.80	**72.27**	23.43
CycleGAN	39.93	54.82	53.76	46.19	65.44	27.85	55.82	40.31	3.94	61.33	83.00	42.17	70.66	57.88	13.88
DiGA	46.88	60.56	59.82	51.32	78.15	31.51	57.84	48.62	9.14	72.59	81.01	57.26	69.29	58.35	35.97
ProDA	38.85	52.70	53.99	49.81	50.53	14.85	58.48	36.92	22.53	66.50	67.13	25.85	73.80	53.93	36.77
MASN	48.86	62.97	50.39	**65.20**	**83.39**	**51.57**	33.36	43.45	25.38	76.32	82.65	58.17	74.61	61.54	29.61
CaGAN	45.35	59.76	46.43	62.23	70.44	31.52	55.46	47.31	11.63	71.58	82.66	47.93	71.35	64.22	20.83
RCA-DD	44.31	58.81	58.78	46.39	68.48	32.75	58.14	49.32	10.55	63.38	81.29	49.34	73.79	65.95	19.09
Our	49.31±1.51	63.50±2.21	63.96±2.09	61.31	75.21	43.87	58.61	27.32	**28.31**	76.57	86.12	61.45	73.36	43.87	**44.32**
Our+CycleGAN	**51.21**±1.01	**65.12**±1.10	**65.55**±1.12	63.12	80.14	49.98	64.21	28.64	23.14	**77.65**	**89.15**	**66.34**	**78.87**	44.39	38.63

*Note: **Bolded data** represents the best results, underlined data indicates the second-best results.

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TABLE VI
QUANTITATIVE RESULTS OF THE "POT-IRRG → VAI" SCENARIO

Methods	mIoU(%)	mF1(%)	mDice(%)	IoU(%)						F1(%)					
				Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter	Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter
Source-only	31.01	44.33	44.69	36.21	52.12	16.32	55.12	21.57	6.32	53.12	68.12	27.21	71.21	35.31	11.21
MCD	43.08	57.42	57.43	55.20	64.41	25.27	61.85	39.93	11.86	71.13	78.36	40.34	76.43	57.07	21.20
CLAN	43.62	59.11	59.10	51.83	60.19	33.70	60.98	35.41	19.58	68.28	75.15	50.41	75.76	52.30	32.75
CCDA	52.03	66.02	67.84	48.21	76.75	47.02	55.03	44.90	20.71	80.13	86.65	64.16	71.90	61.94	31.34
CycleGAN	36.51	50.12	50.49	50.23	61.21	22.15	58.98	20.45	8.61	67.15	76.89	36.71	73.64	34.54	15.67
DiGA	45.38	61.43	61.49	49.41	62.30	38.87	57.66	34.34	29.66	45.65	76.77	55.97	73.14	51.09	45.65
ProDA	44.61	58.64	58.65	55.31	68.67	32.54	61.01	41.98	8.24	72.25	81.43	49.11	75.78	59.14	15.22
MASN	50.05	63.71	66.11	60.31	69.56	44.64	61.15	**50.14**	21.21	72.65	80.45	60.22	71.52	61.09	42.30
CaGAN	51.27	66.05	66.55	59.42	66.63	42.41	63.84	49.36	25.94	74.53	79.97	59.56	77.93	**66.10**	41.20
RCA-DD	50.20	64.50	65.50	59.50	65.67	43.78	61.31	48.08	22.89	74.61	79.28	60.90	76.01	64.94	37.26
Our	52.31±0.85	66.31±1.42	**68.70**±1.49	65.32	82.32	47.35	62.32	28.35	30.35	79.36	90.12	64.32	77.35	44.31	46.31
Our+CycleGAN	**54.12**±0.78	**68.15**±1.21	68.31±1.30	**68.21**	**83.78**	**52.91**	**64.54**	26.81	**31.25**	**81.02**	**91.32**	**68.78**	**78.52**	41.78	**52.32**

*Note: **Bolded data** represents the best results, underlined data indicates the second-best results.

TABLE VII
QUANTITATIVE RESULTS OF THE "VAI → POT-IRRG" SCENARIO

Methods	mIoU(%)	mF1(%)	mDice(%)	IoU(%)						F1(%)					
				Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter	Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter
Source-only	29.29	44.97	45.75	38.32	38.36	26.21	24.76	29.95	**22.21**	55.36	55.45	41.32	38.74	45.63	**36.21**
MCD	35.37	49.33	50.33	37.21	54.33	32.98	39.44	48.28	3.14	54.24	70.41	49.61	56.57	65.12	5.90
CLAN	35.30	49.10	50.58	34.02	59.90	35.30	37.66	44.93	4.64	50.77	74.93	52.18	54.71	62.00	6.15
CCDA	43.17	57.77	57.82	57.65	65.44	29.77	35.87	56.99	13.27	**73.14**	79.11	45.88	52.80	72.28	23.43
CycleGAN	37.63	52.95	52.97	45.96	59.01	41.73	25.80	39.71	13.56	62.97	74.22	58.87	40.97	56.84	23.84
DiGA	43.99	58.12	55.34	**60.37**	64.85	9.50	48.43	**76.08**	4.74	63.96	78.14	47.35	**72.42**	63.12	25.41
ProDA	36.52	50.63	50.62	35.86	57.60	38.82	42.61	43.26	0.92	52.79	73.10	55.93	59.76	60.39	1.83
MASN	42.14	57.32	59.91	51.51	**77.51**	31.21	**55.35**	48.74	10.98	57.36	57.36	20.64	38.68	**77.63**	10.69
CaGAN	40.74	54.86	54.85	48.97	64.75	38.43	43.52	45.77	3.02	65.75	78.60	55.52	60.65	62.80	5.85
RCA-DD	40.01	54.22	54.20	44.08	57.26	36.79	41.80	57.15	2.96	61.19	72.82	53.79	58.95	72.74	5.82
Our	44.72±0.71	59.82±1.51	60.39 ±1.43	49.65	67.98	49.32	48.57	42.87	13.36	66.14	80.98	66.65	65.32	59.47	23.64
Our+CycleGAN	**46.84**±1.01	**61.23** ±1.71	**62.33** ±1.75	54.35	69.78	**51.36**	50.87	43.36	14.68	70.98	**81.12**	**68.65**	66.32	60.78	25.32

*Note: **Bolded data** represents the best results, underlined data indicates the second-best results.

TABLE VIII
QUANTITATIVE RESULTS OF THE "VAI → POT-RGB" SCENARIO

Methods	mIoU(%)	mF1(%)	mDice(%)	IoU(%)						F1(%)					
				Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter	Impe. Sur.	Buil.	Low Veg.	Tree	Car	Clutter
Source-only	36.01	52.28	53.06	44.35	49.81	43.32	28.78	30.97	23.47	61.78	66.31	60.82	44.21	46.54	37.19
MCD	36.49	50.95	50.94	39.59	54.58	30.19	41.51	49.27	3.80	56.72	70.62	46.37	58.67	66.02	7.32
CLAN	37.28	50.67	51.61	55.18	43.48	43.07	23.97	58.00	2.91	71.17	60.60	60.21	38.67	73.42	1.21
CCDA	46.87	61.66	62.21	**64.39**	66.44	47.17	37.55	59.35	12.31	78.59	79.84	63.27	54.60	75.08	24.59
CycleGAN	39.30	54.71	54.72	51.01	53.37	36.50	34.98	48.49	11.48	67.53	69.59	53.48	51.82	65.31	20.56
DiGA	45.91	59.30	59.50	58.35	68.12	54.89	53.15	40.32	2.32	73.65	81.32	70.32	69.63	58.67	5.63
ProDA	37.76	51.29	51.28	32.91	62.99	33.93	41.02	55.31	0.43	49.53	77.29	50.67	58.19	71.22	0.85
MASN	49.12	63.15	64.30	61.65	69.51	43.31	**60.15**	44.83	17.69	**79.70**	77.84	67.58	64.06	76.51	16.15
CaGAN	46.55	59.16	59.17	62.51	65.75	49.64	33.21	**67.10**	1.10	76.93	79.33	66.34	49.86	49.86	2.17
RCA-DD	46.36	59.12	59.11	60.38	64.02	49.00	38.58	65.99	0.21	75.29	78.06	65.78	55.68	**79.51**	0.41
Our	49.32±1.53	63.32±2.31	66.05±2.52	54.35	67.25	53.45	51.64	42.35	**31.74**	70.65	80.14	69.45	68.65	59.87	**47.32**
Our+CycleGAN	**51.42** ±1.21	**66.36** ±2.12	**67.35** ±2.14	58.32	**69.74**	**57.54**	56.69	46.98	24.21	73.12	**82.65**	**72.14**	**72.69**	63.54	38.31

*Note: **Bolded data** represents the best results, underlined data indicates the second-best results.

Although the proposed NAPG model is similar to the MASN model, it still faces challenges in dealing with smaller or more complex categories due to the limitations of prototype representation. Overall, the experimental results show considerable performance differences on different datasets, highlighting the impact of data distribution differences between the source and target domains on the performance of UDA models.

During the experimental evaluation of the proposed NAPG model, the performance on some land cover categories did not reach the optimal level. Specifically, as shown in Table V, the IoU of the "car" is significantly lower than that of the CCDA model. The reason is that the internal distribution of the "car" is relatively complex, which makes it difficult for the MPG module in the model to fully and comprehensively capture the features of this category, thereby adversely affecting its segmentation effect. At the same time, the feature distribution differences between the source and target domains also weaken the cross-domain alignment effect to a certain extent. As shown in Table VII, the IoU of the proposed method on the "clutter" category is only 14.68, lower than other methods, such as the DiGA method's 29.66. This indicates that the combined effect of the distribution complexity of this category and the limitations of cross-domain feature alignment has led to the proposed method not achieving the desired optimal results. In some cases, there may be confusion between the features of different categories, which brings difficulties to the model's discrimination work. For example, in some scenes,

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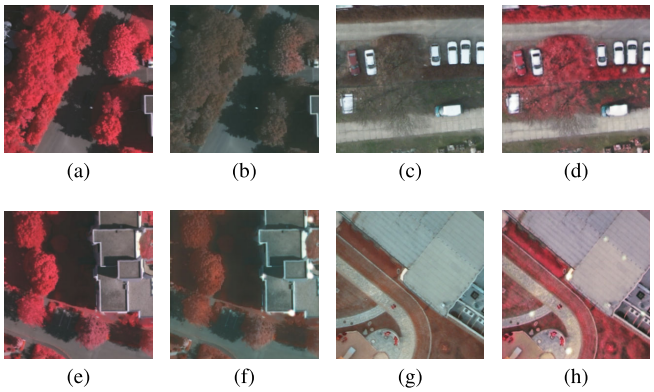


Fig. 15. Visual style transfer across different data domains using the CycleGAN model [65]. The sample style of the source domain is transformed to match the sample style of the target domain, increasing visual similarity and facilitating better UDA. (a) Real VAI sample. (b) Generated POT-RGB sample. (c) Real POT-RGB sample. (d) Generated VAI sample. (e) Real VAI sample. (f) Generated POT-IRRG sample. (g) Real POT-IRRG sample. (h) Generated VAI sample.

the "Low Veg." and Tree categories may have similar spectral features, which makes the NAPG model prone to confusion when performing segmentation operations.

However, from the overall results of the comparative experiments based on the mIoU and mF1 parameters, the NAPG model still shows its unique advantages and achieves the optimal overall effect, especially in the recognition of the "Buil." and "Low Veg." categories. The overall performance can be primarily attributed to the proposed MPG module, which adaptively generates prototype features to effectively represent and separate distinct land cover classes. Furthermore, the GNCE module contributes significantly by enhancing the intraclass consistency, particularly for large-area land cover classes.

As discussed in Section II, incorporating style transfer techniques is a widely used and effective strategy. To explore the potential applications of the NAPG model, the CycleGAN-based [65] style transfer algorithm was used to transform the style of the source-domain data to the target-domain style. The generated data were added to the source-domain dataset for supervised training. By training with labeled data similar to the target-domain style, the model gradually improved its ability to fit the target-domain distribution, thereby enhancing its generalization capabilities. Specifically, as shown in Tables V through VIII, the integration of CycleGAN improved the algorithm's mIoU performance by approximately 4%. In addition, as shown in Fig. 15, the visualizations of the images before and after style transfer show improved visual similarity between the generated images and the target domain. These results indicate that CycleGAN effectively simulates the data distribution of the target domain to a certain extent. To more comprehensively verify the effectiveness of the proposed method, a UDA RS-SS performance comparison experiment on the LoveDA dataset [66] has been added in Appendix B.

V. Conclusion

This article presents the NAPG model for UDA in RS-SS. The model addresses the challenges of complex land cover class distributions and limited spatial information in prototype-based UDA methods. The NAPG model integrates two novel modules into the DeepLab-V3 semantic segmentation model: 1) multiprototype group-based cross-domain representation (CDR-MPG): this module generates multiprototype feature groups to improve the model's ability to represent different land cover classes; and 2) gradient neighborhood consistency estimation (GNCE): this module improves the prototype space representation by estimating gradient similarity within neighborhoods and applying weighted constraints. Based on the experimental results, the CDR-MPG module outperforms single/multiprototype features of equal number when handling complex class distributions. In the GNCE module, we explore the influence of different neighborhood ranges on model performance, analyzing and validating performance improvements over different land cover objects. The model is also compared with several state-of-the-art methods, demonstrating its superiority in semantic segmentation UDA tasks for remote sensing images. While generally robust, NAPG shows limitations with specific small or highly complex classes where the CDR-MPG module struggles with subtle features amidst significant distributional complexity. Future work will explore improved prototype determination within groups and dynamic/learnable approaches to enhance CDR-MPG's representational ability.

TABLE IX
Overview of Experimental Datasets

Name	Band	Resolution	Size	Number
LoveDA-Urban	R-G-B	0.3 m/pixel	1024×1024	2993
LoveDA-Rural	R-G-B	0.3 m/pixel	1024×1024	2994

*Note: R: Red band. G: Green band. B: Blue band.

TABLE X
Proportion of Prototypes in the Normalized MPG

Data	Number of prototypes
LoveDA-Urban → LoveDA-Rural	4,2,1,1,1,1,1
LoveDA-Rural → LoveDA-Urban	4,1,1,1,1,4,2

Appendix

A. Prototype Similarity Cross Entropy Optimization Function

Traditionally, the prototype optimization function based on Euclidean distance is defined as

$$\mathcal{L}_2 = \sum_{i,j}\sum_{c} \mathcal{I}_{l,i,j,c}\left\|\mathcal{P} - \mathcal{F}_{i,j}\right\|^2 \tag{20}$$

where $\mathcal{I}_{l,i,j,c}$ represents the source-domain label or the pseudo-label of the target domain, $\mathcal{F}_{i,j}$ represents the deep features of a specific sample, and $\mathcal{P}$ is the prototype feature. The derivative of $\mathcal{L}_2$ with respect to $\mathcal{F}$ is obtained as

$$\begin{aligned}\frac{\partial \mathcal{L}_2}{\partial \mathcal{F}} &= -2\sum_{i,j}\sum_{c} \mathcal{I}_{l,i,j,c}\mathcal{P} + 2\sum_{i,j}\sum_{c} \mathcal{I}_{l,i,j,c}\mathcal{F} \\ &= 2\sum_{i,j}\sum_{c} \mathcal{I}_{l,i,j,c}(\mathcal{F} - \mathcal{P}).\end{aligned} \tag{21}$$

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TABLE XI
QUANTITATIVE RESULTS OF THE LOVEDA-URBAN → LOVEDA-RURAL SCENARIO

Method	mIoU(%)	mF1(%)	mDice(%)	IoU(%)							F1(%)						
				Bg.	Buil.	Rd.	Wat.	Barr.	Fst.	Agri.	Bg.	Buil.	Rd.	Wat.	Barr.	Fst.	Agri.
Source-only	29.48	43.11	43.53	50.12	27.65	21.24	58.85	21.67	18.47	10.31	67.32	42.87	35.47	74.64	35.69	30.54	18.15
MCD	30.25	43.63	45.25	37.51	36.89	30.02	53.09	22.78	21.21	11.51	54.61	53.30	46.71	69.15	37.81	34.15	19.87
CLAN	32.61	47.28	45.24	34.01	38.25	34.71	64.09	30.02	13.82	16.09	51.98	55.78	50.54	78.98	46.21	23.15	27.41
CCDA	33.44	48.29	48.52	**52.23**	32.12	28.78	60.12	23.11	19.87	19.02	**69.45**	49.53	44.05	75.36	38.21	31.54	31.39
cycleGAN	32.45	47.97	48.42	30.05	39.78	35.48	54.90	32.73	19.16	18.43	46.95	56.08	52.31	70.46	48.64	32.29	30.15
DiGA	35.44	48.47	49.05	37.15	29.18	21.89	54.17	17.94	**55.61**	21.87	54.69	45.79	35.69	70.60	29.37	**71.69**	35.15
proDA	32.15	47.50	48.04	45.21	35.78	33.35	49.09	27.87	22.15	14.65	62.21	52.97	49.05	66.67	42.08	37.60	24.67
MASN	34.71	50.25	51.07	38.68	27.69	21.54	54.06	18.14	34.10	**55.21**	55.64	42.84	35.61	70.08	30.36	68.57	**51.48**
CaGAN	34.71	49.05	49.99	41.52	34.58	**36.91**	58.74	33.54	30.25	8.29	58.65	51.27	**53.36**	74.58	50.08	46.72	15.49
RCA-DD	33.48	48.81	49.32	46.15	**42.92**	35.08	47.69	28.64	19.78	16.25	63.97	**60.70**	52.47	64.06	44.72	31.55	28.49
Our	36.54±0.89	51.38±1.22	52.01±1.98	47.58	31.29	33.54	**68.77**	36.51	32.67	9.58	64.05	48.26	50.37	**81.46**	53.94	48.02	16.54
Our+GAN	**37.15**±1.25	**52.36**±1.89	**52.55**±2.08	47.52	31.26	32.56	66.59	**40.68**	34.52	9.64	64.22	47.52	49.81	80.34	**57.85**	50.37	17.16

*Note: **Bolded data** represents the best results, underlined data indicates the second-best results.

TABLE XII
QUANTITATIVE RESULTS OF THE LOVEDA-RURAL → LOVEDA-URBAN SCENARIO

Method	mIoU(%)	mF1(%)	mDice(%)	IoU(%)							F1(%)						
				Bg.	Buil.	Rd.	Wat.	Barr.	Fst.	Agri.	Bg.	Buil.	Rd.	Wat.	Barr.	Fst.	Agri.
Source-only	32.81	47.71	48.41	38.12	25.54	23.36	54.78	11.36	49.87	31.01	55.21	40.54	37.12	70.98	20.54	65.46	48.65
MCD	35.31	50.97	51.48	38.21	35.45	23.78	51.45	13.65	48.78	39.65	55.45	51.87	37.85	68.73	24.19	64.14	56.14
CLAN	33.51	48.71	49.32	27.56	31.45	21.85	50.72	13.64	44.98	48.12	43.28	48.69	35.14	67.45	24.98	61.25	64.28
CCDA	37.15	52.64	53.19	39.97	37.12	24.56	56.89	15.14	54.87	35.10	56.95	54.08	39.47	72.15	26.14	70.89	52.04
cycleGAN	33.81	49.71	50.11	35.25	33.36	19.28	45.61	22.06	48.57	35.30	52.08	49.47	31.54	62.67	36.52	65.41	52.87
DiGA	38.21	54.19	54.56	40.12	33.34	28.87	56.69	19.01	55.08	37.49	57.14	49.85	43.34	72.21	32.72	71.35	54.79
proDA	36.33	52.01	52.11	39.01	33.84	20.15	51.07	18.19	53.69	39.14	56.15	49.14	34.87	68.98	31.97	70.09	56.81
MASN	41.13	57.15	57.66	38.12	42.65	28.78	56.14	20.71	54.66	50.36	55.38	59.05	44.82	72.64	33.98	70.15	67.74
CaGAN	40.02	56.08	56.57	30.25	42.87	26.05	53.56	24.45	52.31	54.39	46.16	60.09	41.56	69.18	38.34	68.72	70.34
RCA-DD	38.91	55.15	55.75	28.69	**43.78**	**29.15**	45.25	24.87	51.64	53.14	44.57	**60.25**	**45.58**	62.47	38.51	68.25	69.51
Our	42.25±0.56	57.61±0.84	58.15±1.24	**45.05**	27.29	24.15	55.78	**26.80**	**67.18**	52.61	**62.16**	42.07	39.43	71.05	41.16	**80.63**	69.61
Our+GAN	**42.51**±1.25	**57.86**±1.69	**58.25**±1.87	40.64	25.61	25.15	**60.64**	26.71	65.25	**56.75**	57.06	40.04	40.92	**75.64**	**42.69**	79.65	**72.36**

*Note: **Bolded data** represents the best results, underlined data indicates the second-best results.

Similarly, based on (7), the derivative of the multiprototype similarity CE loss $\mathcal{L}_{CE}^{\text{si}}$ with respect to the similarity map $\mathcal{SI}_c$ is derived as

$$\begin{aligned}
\frac{\partial \mathcal{L}_{\text{CE}}^{\text{si}}}{\partial \mathcal{SI}_c} &= \frac{\partial}{\partial \mathcal{SI}_c}\left(-\sum_{i,j}\sum_{c}\mathcal{I}_{l,i,j,c}\log\left(\hat{y}_{i,j,c}^{\text{si}}\right)\right) \\
&= -\sum_{c}\mathcal{I}_{l,i,j,c}\frac{\partial \log\left(\hat{y}_{i,j,c}^{\text{si}}\right)}{\partial \mathcal{SI}_{i,j,c}} \\
\frac{\partial \hat{y}_{i,j,c}^{\text{si}}}{\partial \mathcal{SI}_c} &= \hat{y}_{i,j,c}^{\text{si}}\left(\vartheta_{c,c'} - \hat{y}_{i,j,c}^{\text{si}}\right) \\
\frac{\partial \mathcal{L}_{\text{CE}}^{\text{si}}}{\partial \mathcal{SI}_c} &= \hat{y}_{i,j,c}^{\text{si}} - \mathcal{I}_{l,i,j,c}
\end{aligned} \tag{22}$$

where $\hat{y}_{i,j,c}^{\text{si}}$ represents the similarity probability obtained through SoftMax normalization. The Kronecker delta $\vartheta_{c,c'}$ is defined as $\vartheta_{c,c'} = 1$ when $c = c'$ and $\vartheta_{c,c'} = 0$ otherwise. $\mathcal{I}_{l,i,j,c}$ denotes the label of the sample, where 1 indicates the sample belongs to class c and 0 otherwise. The gradient $\hat{y}_{i,j,c}^{\text{si}} - \mathcal{I}_{l,i,j,c}$ reflects the difference between the predicted probability and the true label, guiding adjustments to deep features to better represent their respective classes. This process ensures that the deep features align more closely with the prototype features, improving classification accuracy.

By taking the partial derivative of $\mathcal{L}_{\text{CE}}^{\text{si}}$ with respect to the deep feature $\mathcal{F}$ and using (22) and (2), the following expression is derived:

$$\begin{aligned}
\frac{\partial \mathcal{L}_{\text{CE}}^{\text{si}}}{\partial \mathcal{F}} &= \sum_{c}\frac{\partial \mathcal{L}_{\text{CE}}^{\text{si}}}{\partial \mathcal{SI}_c}\cdot\frac{\partial \mathcal{SI}_c}{\partial \mathcal{F}} \\
\frac{\partial \mathcal{SI}_c}{\partial \mathcal{F}} &= \mathcal{P}_c \\
\frac{\partial \mathcal{L}_{\text{CE}}^{\text{si}}}{\partial \mathcal{F}} &= \sum_{c}\left(\hat{p}_{i,j,c}^{\text{si}} - \mathcal{I}_{l,i,j,c}\right)\cdot\mathcal{P}_c \\
\frac{\partial \mathcal{L}_{\text{CE}}^{\text{si}}}{\partial \mathcal{F}} &= \sum_{c}\left(\frac{\exp(\langle \mathcal{P}_c, \mathcal{F}\rangle)}{\sum_c \exp(\langle \mathcal{P}_c, \mathcal{F}\rangle)} - \mathcal{I}_{l,i,j,c}\right)\cdot\mathcal{P}_c.
\end{aligned} \tag{23}$$

Based on the above, the similarity CE loss not only considers the distance between features but also incorporates a directional constraint through the inner product $\langle \mathcal{P}_c, \mathcal{F}\rangle$. This dual constraint effectively guides the deep feature $\mathcal{F}$ to progressively align with the prototype feature $\mathcal{P}$, while preserving the relative similarity among different classes. Consequently, the model's ability to express both interclass and intraclass feature relationships is significantly enhanced.

B. Performance Comparison Supplementary Experiment

To more comprehensively verify the effectiveness of the proposed method, a comparison experiment on UDA performance with advanced methods was conducted on the LoveDA remote sensing dataset [66]. The LoveDA dataset contains data in both urban and rural styles, both of which follow a unified labeling rule. The dataset includes seven types of land cover classes, namely, background (Bg.), building (Buil.), road (Rd.), water

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(Wat.), barren (Barr.), forest (Fst.), and agriculture (Agri.). Their detailed properties are shown in Table IX.

The way the LoveDA dataset [66] constructs the data for UDA experiments in the field of RS-SS is as follows.

1) *LoveDA-Urban → LoveDA-Rural:* The source domain and target domain are the urban and rural style data in the LoveDA dataset, respectively. This task involves domains that are different in terms of imaging location.
2) *LoveDA-Rural → LoveDA-Urban*: The source domain and target domain are the rural and urban style data in the LoveDA dataset, respectively. This task involves domains that are different in terms of imaging location.

After the warm-up phase, the η_c values for different categories in the two datasets [normalized and rounded according to (6)] are shown in Table X. It can be seen from the table that there is a significant difference in the richness of land cover types between the urban and rural datasets. For example, the category richness of "Buil." is higher in the Urban dataset, while the category richness of "Agri." is more prominent in the rural dataset. The comparative performance experimental results on the LoveDA dataset are shown in Tables XI and XII, respectively.

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